

SOC/L1 - 6.3. Wireshark Packet Operations

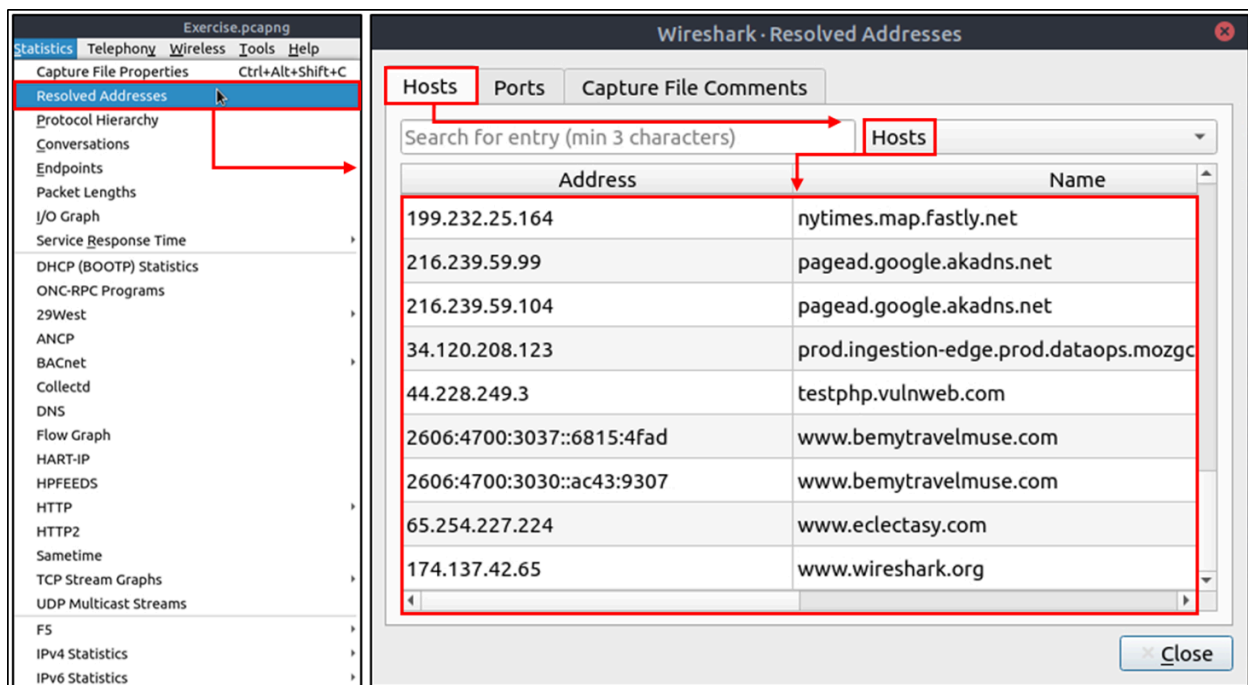
Task 1 - Introduction

In this room, we will cover the fundamentals of packet analysis with Wireshark and investigate the event of interest at the packet-level.

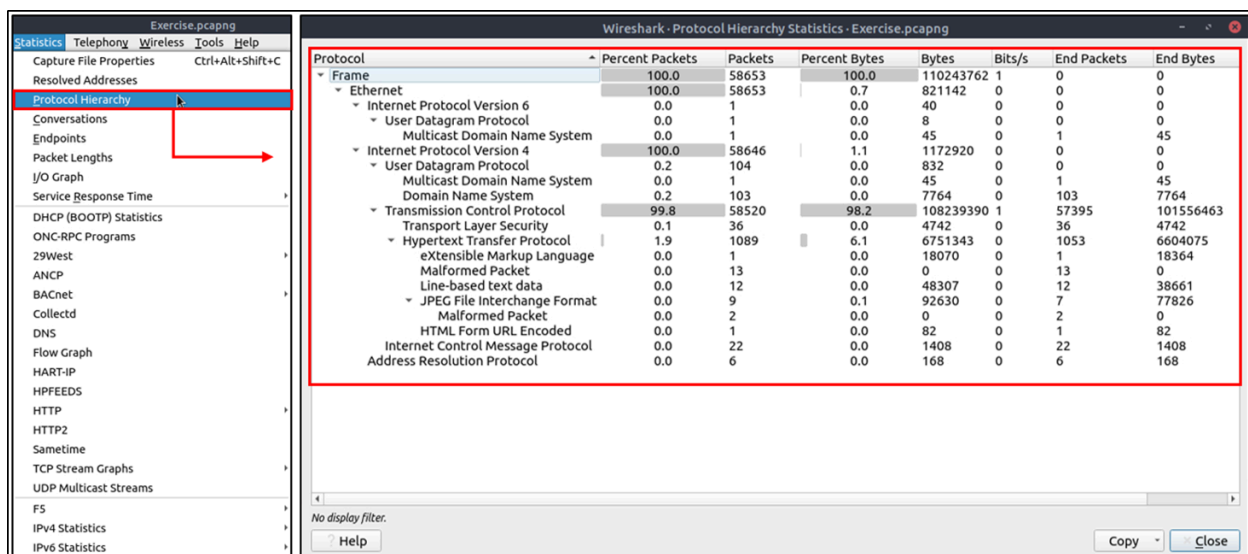
Task 2 - Statistics | Summary

Statistics: Wireshark's **Statistics** menu provides a quick overview of captured network traffic, including protocols, endpoints, conversations, and protocol-specific details such as DNS and DHCP. Security analysts use these statistics to understand traffic patterns, identify suspicious activity, and create investigation hypotheses before performing deeper packet analysis.

Resolved Addresses: Wireshark's **Resolved Addresses** feature displays IP addresses and associated hostnames identified from DNS responses within the PCAP file. Analysts can use this information to quickly identify accessed resources, investigate suspicious domains, and evaluate network activity during security investigations. "**Statistics → Resolved Addresses**" menu to view all resolved addresses by Wireshark.



Protocol Hierarchy: Provides a breakdown of protocols, packet counts, and traffic percentages, helping analysts understand network activity, identify unusual services, and filter relevant traffic for investigation. "Statistics → Protocol Hierarchy"



Conversations: Displays traffic exchanges between endpoints across Ethernet, IPv4, IPv6, TCP, and UDP, helping analysts identify communication relationships and investigate relevant network connections.

Wireshark - Conversations · Exercise.pcapng

Ethernet · 8 **IPv4 · 18** IPv6 · 1 TCP · 19 UDP · 53

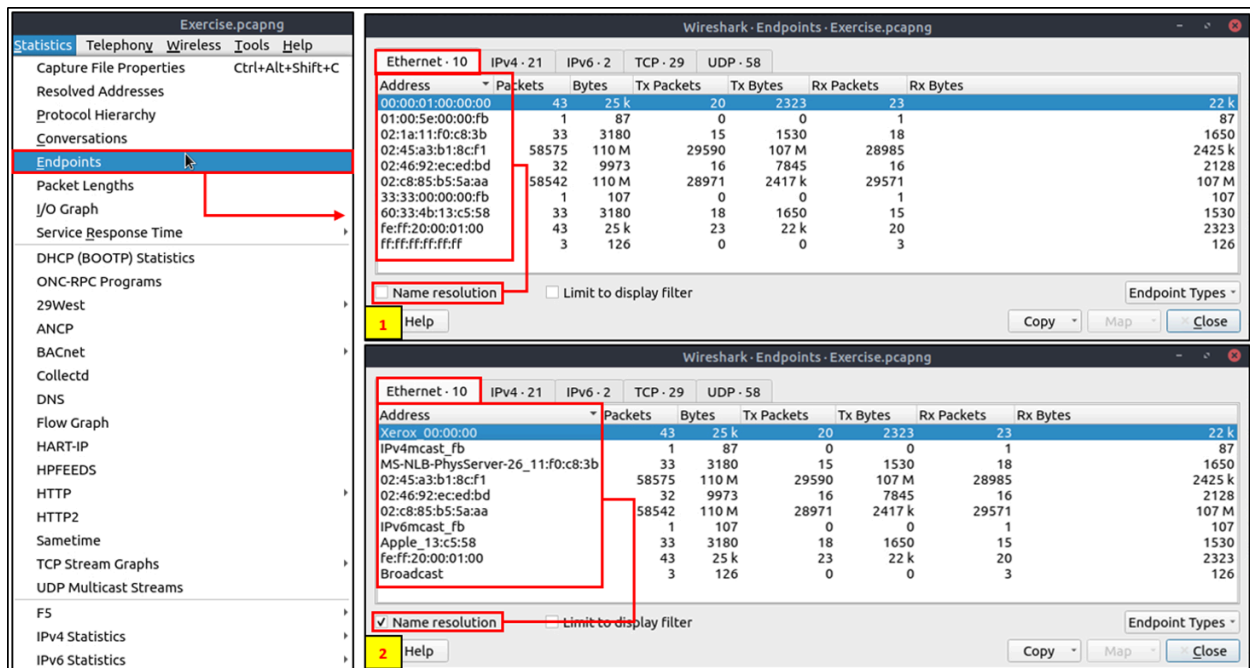
Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start
4.2.2.2	192.168.43.9	6	588	3	294	3	294	510497
8.8.4.4	192.168.43.9	4	392	1	98	3	294	510492
8.8.8.8	192.168.43.9	6	588	3	294	3	294	510489
10.0.0.2	10.10.57.178	90	10 k	45	6323	45	4466	3415438
10.10.47.123	10.10.57.178	31	9931	15	7803	16	2128	3415439
10.10.57.178	10.100.1.33	58194	110 M	29387	107 M	28807	2305 k	3415437
10.10.57.178	34.120.208.123	23	3673	12	2628	11	1045	3415439
10.10.57.178	34.117.237.239	28	4152	16	2291	12	1861	3415467
10.10.57.178	52.43.127.64	4	330	2	167	2	163	3415469
10.10.57.178	224.0.0.251	1	87	1	87	0	0	3415487
10.10.57.178	35.244.181.201	6	473	4	341	2	132	3415488
10.10.57.178	34.120.237.76	7	618	4	342	3	276	3415492
10.10.57.178	44.228.249.3	186	114 k	99	12 k	87	101 k	3415496
65.208.228.223	145.254.160.237	34	20 k	18	19 k	16	1351	0.0000
145.253.2.203	145.254.160.237	2	277	1	188	1	89	2.5536
145.254.160.237	216.239.59.99	7	4119	3	883	4	3236	2.9842
174.137.42.65	192.168.43.9	6	588	3	294	3	294	510500
192.168.43.1	192.168.43.9	11	1024	5	550	6	474	510484

Wireshark - Conversations · Exercise.pcapng

Ethernet · 8 IPv4 · 18 IPv6 · 1 **TCP · 19** UDP · 53

Address A	Port A	Address B	Port B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes
10.10.57.178	54052	10.10.47.123	9696	10	1518	5	687	5	
10.10.57.178	55588	34.120.208.123	443	23	3673	12	2628	11	
10.10.57.178	54054	10.10.47.123	9696	10	1636	5	644	5	
10.10.57.178	54056	10.10.47.123	9696	11	6777	6	797	5	
10.10.57.178	46742	34.117.237.239	443	28	4152	16	2291	12	
10.10.57.178	39766	52.43.127.64	443	4	330	2	167	2	
10.10.57.178	56496	35.244.181.201	443	6	473	4	341	2	
10.10.57.178	48564	34.120.237.76	443	7	618	4	342	3	
10.10.57.178	57672	44.228.249.3	80	43	31 k	23	3898	20	
10.10.57.178	57674	44.228.249.3	80	6	412	4	272	2	
10.10.57.178	57676	44.228.249.3	80	19	5185	10	1033	9	
10.10.57.178	57678	44.228.249.3	80	23	11 k	12	1165	11	
10.10.57.178	57680	44.228.249.3	80	25	16 k	13	1231	12	
10.10.57.178	57682	44.228.249.3	80	25	16 k	13	1231	12	
10.10.57.178	57684	44.228.249.3	80	45	32 k	24	3865	21	
10.100.1.33	43514	10.10.57.178	80	30111	55 M	14711	1180 k	15400	
10.100.1.33	48924	10.10.57.178	80	28083	54 M	14096	1125 k	13987	
145.254.160.237	3372	65.208.228.223	80	34	20 k	16	1351	18	
145.254.160.237	3371	216.239.59.99	80	7	4119	3	883	4	

Endpoints: Shows unique network devices and addresses (Ethernet, IPv4, IPv6, TCP, and UDP) within a capture file, helping analysts identify hosts of interest and optionally resolve MAC addresses to manufacturer names for easier analysis. "Statistics → Endpoints"



If you want to use Name resolution functionalities, you need to activate them through the **"Edit → Preferences → Name Resolution"** menu.

The image shows the Wireshark Preferences dialog with the Name Resolution tab selected. The 'Name Resolution' section has the following settings:

- ☒ Resolve MAC addresses
- ☐ Resolve transport names
- ☐ Resolve network (IP) addresses
- ☒ Use captured DNS packet data for address resolution
- ☒ Use an external network name resolver
- ☐ Use custom list of DNS servers for name resolution
- DNS Servers: Edit...
- Maximum concurrent requests: 500
- ☐ Only use the profile "hosts" file
- ☐ Resolve VLAN IDs
- ☐ Resolve SS7 PCs
- ☐ Enable OID resolution
- ☐ Suppress SMI errors

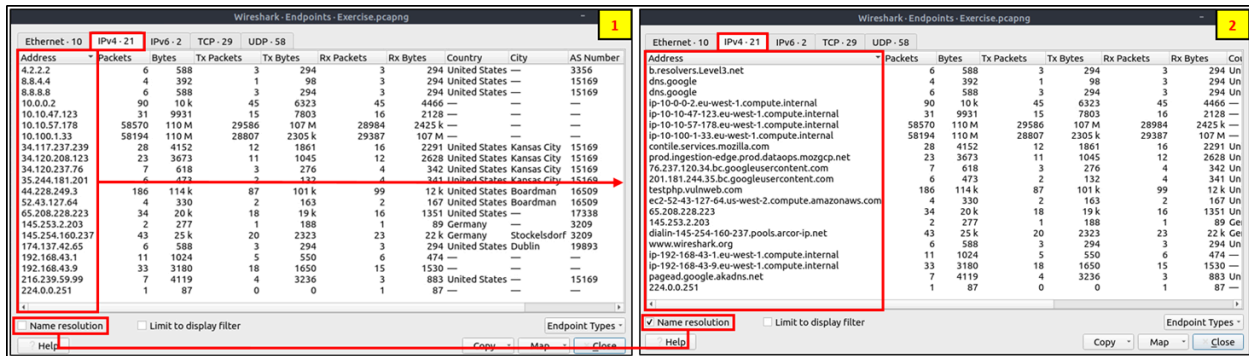
Below the preferences, two screenshots of the packet capture interface are shown. The first screenshot, labeled 'Default installation', shows the packet list with the following data:

Time	Source	Destination	Protocol	Length	Info
1 0.000000	145.254.160.237	65.208.228.223	TCP	62	3372 → 80 [SYN]
2 0.911310	65.208.228.223	145.254.160.237	TCP	62	80 → 3372 [SYN,
3 0.911310	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK]
4 0.911310	145.254.160.237	65.208.228.223	HTTP	533	GET /download.ht
5 1.472116	65.208.228.223	145.254.160.237	TCP	54	80 → 3372 [ACK]

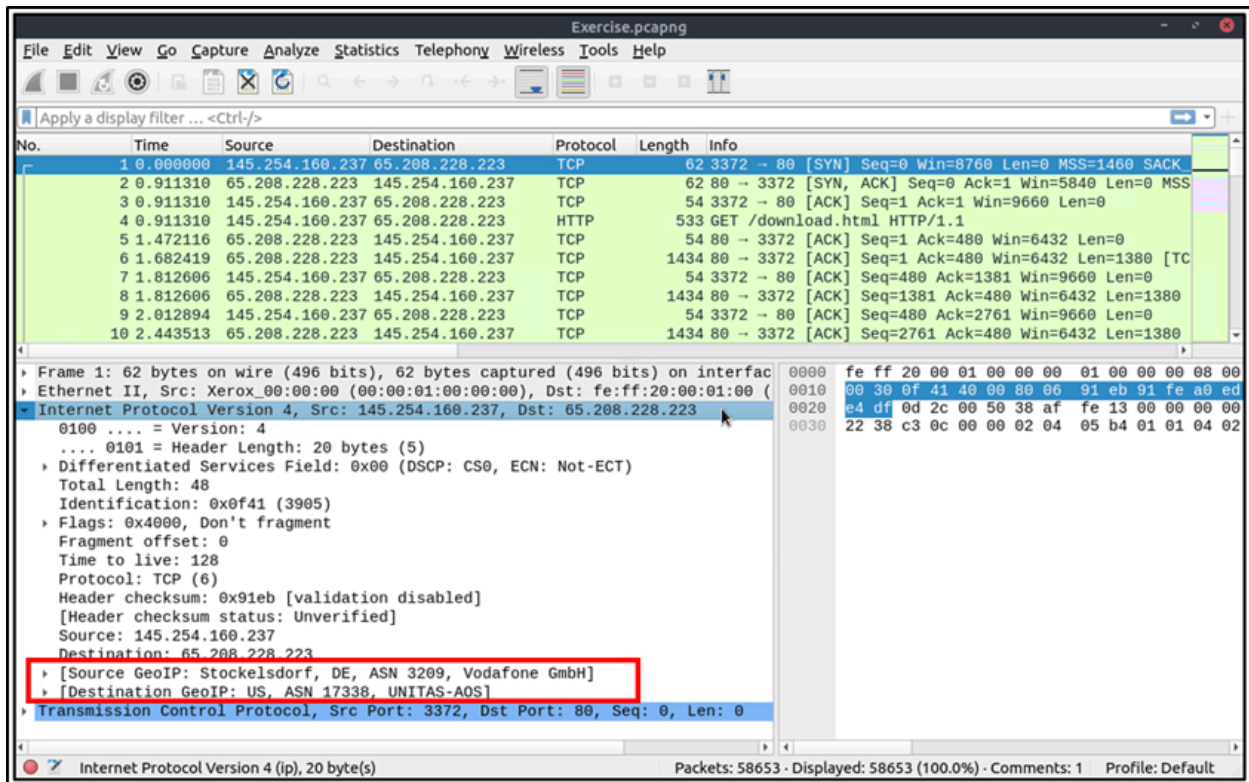
The second screenshot, labeled 'IP and port name resolution', shows the same packet capture with the following data:

Time	Source	Destination	Protocol	Length	Info
1 0.000000	dialin-145-25...	65.208.228.223	TCP	62	tip2(3372) → http(80) [SYN]
2 0.911310	65.208.228.223	dialin-145-254-16...	TCP	62	http(80) → tip2(3372) [SYN,
3 0.911310	dialin-145-25...	65.208.228.223	TCP	54	tip2(3372) → http(80) [ACK]
4 0.911310	dialin-145-25...	65.208.228.223	HTTP	533	GET /download.html HTTP/1.1
5 1.472116	65.208.228.223	dialin-145-254-16...	TCP	54	http(80) → tip2(3372) [ACK]

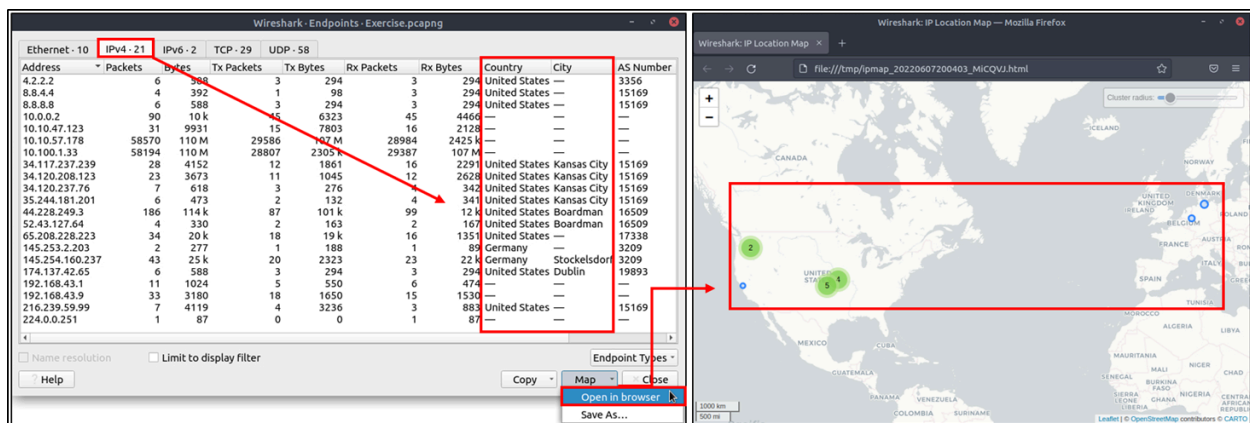
Endpoint menu view with name resolution:



MaxMind DB files and provide the database path to Wireshark by using the "Edit -> Preferences -> Name Resolution -> MaxMind database directories" menu.



Endpoints and GeoIP view:

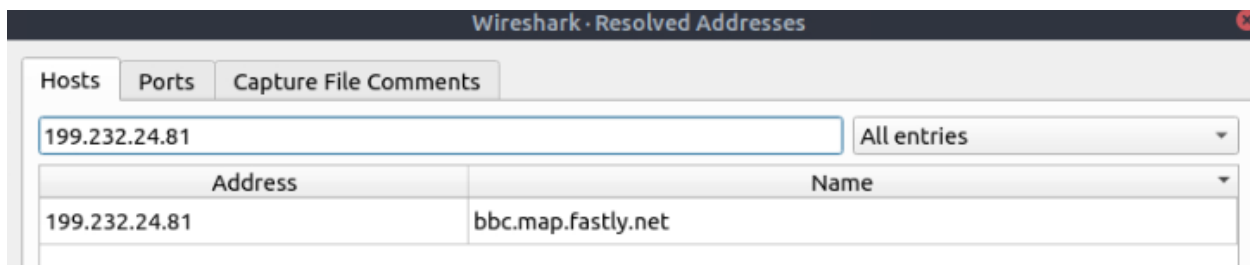


Note: You need an active internet connection to view the GeoIP map. The lab machine doesn't have an active internet connection!

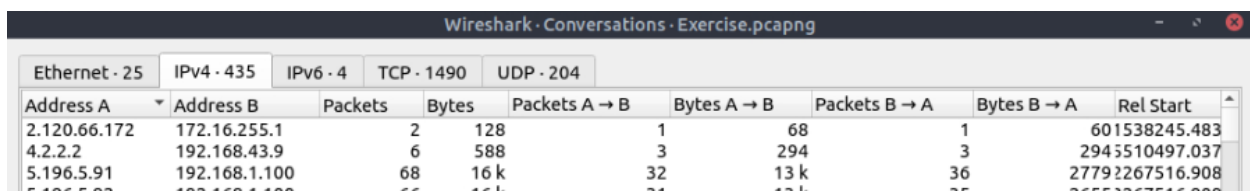
Answer the questions below

Investigate the resolved addresses. What is the IP address of the hostname starts with "bbc"?

199.232.24.81 (Go to the Statistics menu, then click on Resolved Addresses.)



What is the number of IPv4 conversations? 435 (Statics > conversations)



How many bytes (k) were transferred from the "Micro-St" MAC address?

7474 (statistics > endpoints)

Ethernet · 26	IPv4 · 436	IPv6 · 6	TCP · 2164	UDP · 224		
Address	^ Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes
Broadcast	81	18 k	0	0	81	18 k
fe:ff:20:00:01:00	43	25 k	23	22 k	20	2323
zte_2e:f0:5e	8527	831 k	3894	503 k	4633	327 k
LCFCHeFe_ce:28:5f	8571	844 k	4678	341 k	3893	503 k
Apple_13:c5:58	33	3180	18	1650	15	1530
RealtekU_12:35:03	30	2426	0	0	30	2426
RealtekU_12:35:02	1412	519 k	808	411 k	604	108 k
Micro-St_9a:f1:f5	10478	7474 k	4294	1083 k	6184	6390 k
IPv6mcast_fb	1	107	0	0	1	107
IPv6mcast_01	1	174	0	0	1	174
PcsCompu_cc:3f:1b	1444	522 k	636	111 k	808	411 k
02:c8:85:b5:5a:aa	58542	110 M	28971	2417 k	29571	107 M
02:46:92:ec:ed:bd	32	9973	16	7845	16	2128
02:45:a3:b1:8c:f1	58575	110 M	29590	107 M	28985	2425 k
MS-NLB-PhysServer-26_11:f0:c8:3b	33	3180	15	1530	18	1650
MAC-specific-ctrl-ctrl-01	44	2640	0	0	44	2640
IPv4mcast_7f:ff:fa	22	3739	0	0	22	3739
IPv4mcast_fc	4	256	0	0	4	256
IPv4mcast_fb	1	87	0	0	1	87

What is the number of IP addresses linked with "Kansas City"? 4

Ethernet · 26		IPv4 · 436		IPv6 · 6		TCP · 2164		UDP · 224	
ss	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country	City	
9.190.254	2	140	2	140	0	0	Canada	Kamloops	
7.237.239	28	4152	12	1861	16	2291	United States	Kansas City	
0.208.123	23	3673	11	1045	12	2628	United States	Kansas City	
0.237.76	7	618	3	276	4	342	United States	Kansas City	
4.181.201	6	473	2	132	4	341	United States	Kansas City	
05.148.72	2	519	1	60	1	459	United States	Kearney	

Which IP address is linked with "Blicnet" AS Organisation? 188.246.82.7

Ethernet · 26	IPv4 · 436	IPv6 · 6	TCP · 2164	UDP · 224
Address	Country	City	AS Number	AS Organization
74	United States	Asheville	7018	ATT-INTERNET4
60	United States	—	7018	ATT-INTERNET4
300	United States	—	7018	ATT-INTERNET4
90 k	United States	Miami	20940	Akamai International B.V.
1978	United States	Romulus	20940	Akamai International B.V.
419	Hong Kong	Central	45102	Alibaba US Technology Co., Ltd.
531	Germany	Frankfurt am Main	45102	Alibaba US Technology Co., Ltd.
954	Germany	Frankfurt am Main	45102	Alibaba US Technology Co., Ltd.
531	United States	San Mateo	45102	Alibaba US Technology Co., Ltd.
1062	United States	—	45102	Alibaba US Technology Co., Ltd.
954	United States	—	45102	Alibaba US Technology Co., Ltd.
1062	United States	—	45102	Alibaba US Technology Co., Ltd.
4744	France	Pantin	39894	Aufeminin S.A.
60	United States	Birmingham	33363	BHN-33363
197	United States	Cocoa Beach	33363	BHN-33363
534	United States	Lutz	33363	BHN-33363
966	Ireland	—	2110	BT Communications Ireland Limited
1319	Ireland	—	2110	BT Communications Ireland Limited
76	Bosnia and Herzegovina	—	21107	Blicnet d.o.o.
1489	—	—	5400	British Telecommunications PLC
2906	United Kingdom	—	5400	British Telecommunications PLC

Task 3 - Statistics | Protocol Details

IPv4 and IPv6: Provides separate statistics for IPv4 and IPv6 traffic, allowing analysts to filter, identify, and investigate events related to a specific IP version. "Statistics → IPvX Statistics" menu to view this info.

The screenshot displays the Wireshark interface with the 'Statistics' pane on the left and four data panes on the right. Red arrows indicate the navigation path from the 'Statistics' pane to the specific data panes.

Statistics Pane: The 'IPv4 Statistics' option is selected under the 'Statistics' menu.

Wireshark - All Addresses - Exercise.pcapng

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
All Addresses	58646				0.0000	100%	2.2200	568415475.092
8.8.8.8	6	8.8.8.8			0.0000	0.01%	0.0100	285510489.972
8.8.4.4	4	8.8.4.4			0.0000	0.01%	0.0100	285510492.942
65.208.228.223	34	65.208.228.223			0.0000	0.06%	0.0300	0.911
52.43.127.64	4	52.43.127.64			0.0000	0.01%	0.0300	568415469.634
44.228.249.3	186	44.228.249.3			0.0000	0.32%	0.5100	568415507.869
4.2.2.2	6	4.2.2.2			0.0000	0.01%	0.0200	285510497.037
35.244.181.201	6	35.244.181.201			0.0000	0.01%	0.0600	568415488.790
34.120.237.76	7	34.120.237.76			0.0000	0.01%	0.0400	568415492.778
34.120.208.123	23	34.120.208.123			0.0000	0.04%	0.1000	568415440.158
34.117.237.239	28	34.117.237.239			0.0000	0.05%	0.1600	568415467.779
224.0.0.251	1	224.0.0.251			0.0000	0.00%	0.0100	568415487.070
216.239.59.99	7	216.239.59.99			0.0000	0.01%	0.0300	3.916
192.168.43.9	33	192.168.43.9			0.0000	0.06%	0.0300	285510489.960
192.168.43.1	11	192.168.43.1			0.0000	0.02%	0.0200	285510489.960
174.137.42.65	6	174.137.42.65			0.0000	0.01%	0.0100	285510500.825
145.254.160.237	43	145.254.160.237			0.0000	0.07%	0.0400	2.554
145.253.2.203	2	145.253.2.203			0.0000	0.00%	0.0100	2.554
10.100.1.33	58194	10.100.1.33			0.0000	99.23%	2.2200	568415475.092
10.10.57.178	58570	10.10.57.178			0.0000	99.87%	2.2200	568415475.092
10.10.47.123	31	10.10.47.123			0.0000	0.05%	0.1100	568415444.167
10.0.0.2	90	10.0.0.2			0.0000	0.15%	0.1600	568415439.987

Wireshark - IP Protocol Types - Exercise.pcapng

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
IP Protocol Types	58646				0.0000	100%	2.2200	568415475.092
UDP	104				0.0000	0.18%	0.1600	568415439.987
TCP	58520				0.0000	99.79%	2.2200	568415475.092
NONE	22				0.0000	0.04%	0.0200	285510497.037

Wireshark - Destinations and Ports - Exercise.pcapng

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
145.254.160.237	23				0.0000	0.04%	0.0200	2.554
UDP	1				0.0000	4.35%	0.0100	2.914
3009	1				0.0000	100.00%	0.0100	2.914
TCP	22				0.0000	95.65%	0.0200	2.554
3372	18				0.0000	81.82%	0.0200	2.554
3371	4				0.0000	18.18%	0.0200	3.916
145.253.2.203	1				0.0000	0.00%	0.0100	2.554
UDP	1				0.0000	100.00%	0.0100	2.554
53	1				0.0000	100.00%	0.0100	2.554
TCP	29387				0.0000	50.11%	1.1100	568415475.093
48924	13987				0.0000	100.00%	1.1100	568415475.093
43514	15400				0.0000	47.60%	0.8000	568415474.560
10.10.57.178	28984				0.0000	52.40%	0.8300	568415468.711
UDP	45				0.0000	49.42%	1.2100	568415512.371
59621	1				0.0000	0.16%	0.0800	568415439.988
58648	1				0.0000	2.22%	0.0100	568415439.988
58212	1				0.0000	2.22%	0.0100	568415508.025
57199	1				0.0000	2.22%	0.0100	568415439.992
56919	1				0.0000	2.22%	0.0100	568415438.276
56914	1				0.0000	2.22%	0.0100	568415496.706
56712	1				0.0000	2.22%	0.0100	568415520.218
56495	1				0.0000	2.22%	0.0100	568415507.387
54401	1				0.0000	2.22%	0.0100	568415444.331

Wireshark - Source and Destination Addresses - Exercise.pcapng

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
Source IPv4 Addresses	58646				0.0000	100%	2.2200	568415475.092
8.8.8.8	3				0.0000	0.01%	0.0100	285510490.464
8.8.4.4	1				0.0000	0.00%	0.0100	285510494.282
65.208.228.223	18				0.0000	0.03%	0.0200	2.554
52.43.127.64	2				0.0000	0.00%	0.0100	568415469.634
44.228.249.3	87				0.0000	0.15%	0.2800	568415507.869
4.2.2.2	3				0.0000	0.01%	0.0100	285510497.037
35.244.181.201	2				0.0000	0.00%	0.0200	568415488.792
34.120.237.76	3				0.0000	0.01%	0.0200	568415492.780
34.120.208.123	11				0.0000	0.02%	0.0600	568415440.158
34.117.237.239	12				0.0000	0.02%	0.0600	568415467.780
216.239.59.99	4				0.0000	0.01%	0.0200	3.916
192.168.43.9	18				0.0000	0.03%	0.0200	285510489.960
192.168.43.1	5				0.0000	0.01%	0.0100	285510489.960
174.137.42.65	3				0.0000	0.01%	0.0100	285510501.595
145.254.160.237	20				0.0000	0.03%	0.0200	0.911
145.253.2.203	1				0.0000	0.00%	0.0100	2.914
10.100.1.33	28807				0.0000	49.12%	1.2100	568415512.371
10.10.57.178	29586				0.0000	50.45%	1.1100	568415475.092
10.10.47.123	15				0.0000	0.03%	0.0500	568415439.938
10.0.0.2	45				0.0000	0.08%	0.0800	568415439.988
Destination IPv4 Addresses	58646				0.0000	100%	2.2200	568415475.092
8.8.8.8	3				0.0000	0.01%	0.0100	285510489.972
8.8.4.4	3				0.0000	0.01%	0.0100	285510492.942
65.208.228.223	16				0.0000	0.03%	0.0200	0.911
52.43.127.64	2				0.0000	0.00%	0.0200	568415469.634
44.228.249.3	99				0.0000	0.17%	0.2300	568415507.870
4.2.2.2	3				0.0000	0.01%	0.0100	285510497.037
35.244.181.201	4				0.0000	0.01%	0.0400	568415488.790
34.120.237.76	4				0.0000	0.01%	0.0200	568415492.778
34.120.208.123	12				0.0000	0.02%	0.0600	568415440.158
34.117.237.239	16				0.0000	0.03%	0.1000	568415467.779
224.0.0.251	1				0.0000	0.00%	0.0100	568415487.070
216.239.59.99	3				0.0000	0.01%	0.0100	3.916

DNS: Provides a detailed breakdown of DNS traffic, including queries, response codes, query types, and usage statistics, helping analysts identify DNS activity patterns and investigate suspicious communications. "Statistics → DNS"

Exercise.pcapng		Wireshark - DNS - Exercise.pcapng											
Statistics	Telephony	Wireless	Tools	Help	Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst
Capture File Properties	Ctrl+Alt+Shift+C				▼ Total Packets	103				0.0000	100%	0.1600	5684
Resolved Addresses					▼ rcode	103				0.0000	100.00%	0.1600	5684
Protocol Hierarchy					No error	103				0.0000	100.00%	0.1600	5684
Conversations					▼ opcodes	103				0.0000	100.00%	0.1600	5684
Endpoints					Standard query	103				0.0000	100.00%	0.1600	5684
Packet Lengths					▼ Query/Response	103				0.0000	100.00%	0.1600	5684
I/O Graph					Response	51				0.0000	49.51%	0.0800	5684
Service Response Time					Query	52				0.0000	50.49%	0.0800	5684
DHCP (BOOTP) Statistics					▼ Query Type	103				0.0000	100.00%	0.1600	5684
ONC-RPC Programs					PTR (domain name PoinTeR)	7				0.0000	6.80%	0.0200	2855
29West					AAAA (IPv6 Address)	62				0.0000	60.19%	0.1200	5684
ANCP					A (Host Address)	34				0.0000	33.01%	0.0400	5684
BACnet					▼ Class	103				0.0000	100.00%	0.1600	5684
Collectd					IN	103				0.0000	100.00%	0.1600	5684
Flow Graph					▼ Service Stats	0				0.0000	100%	-	-
HART-IP					request-response time (secs)	51	0.03	0.000075	0.413905	0.0000	-	0.0800	5684
HPFEEDS					no. of unsolicited responses	0				0.0000	-	-	-
HTTP					no. of retransmissions	0				0.0000	-	-	-
HTTP2					▼ Response Stats	0				0.0000	100%	-	-
Sametime					no. of questions	102	1.00	1	1	0.0000	-	0.1600	5684
TCP Stream Graphs					no. of authorities	102	0.22	0	1	0.0000	-	0.1600	5684
UDP Multicast Streams					no. of answers	102	0.98	0	8	0.0000	-	0.1600	5684
F5					no. of additional	102	0.88	0	1	0.0000	-	0.1600	5684
IPv4 Statistics					▼ Query Stats	0				0.0000	100%	-	-
IPv6 Statistics					Qname Len	52	27.19	9	55	0.0000	-	0.0800	5684
					▼ Label Stats	0				0.0000	-	-	-
					4th Level or more	31				0.0000	-	0.0800	5684
					3rd Level	20				0.0000	-	0.0400	5684
					2nd Level	1				0.0000	-	0.0100	5684
					1st Level	0				0.0000	-	-	-
					Payload size	103	75.38	35	282	0.0000	100%	0.1600	5684
					Display filter:						Apply		
										Copy	Save as...	Close	

HTTP: Provides a breakdown of HTTP traffic, including request/response codes and original requests, helping analysts understand web activity and investigate potential security events. "Statistics → HTTP"

The image displays the Wireshark interface for the file 'Exercise.pcapng'. The main packet list shows a series of HTTP packets. The Statistics panel on the left has 'HTTP' selected. The Packet Counter, Requests, Load Distribution, and Request Sequences panels are all open and highlighted with red boxes. The Load Distribution panel shows HTTP Requests by Server Address. The Request Sequences panel shows a list of HTTP requests to various URLs.

Answer the questions below

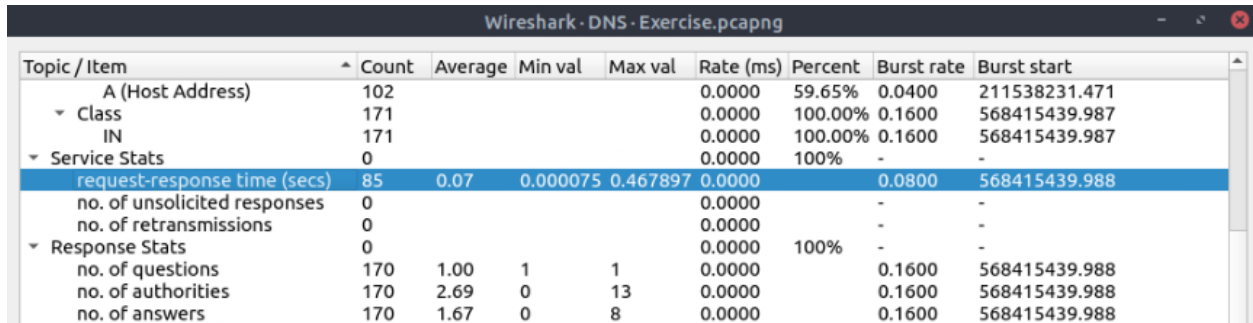
What is the most used IPv4 destination address?

10.100.1.33 (Statistics > IPv4 statistics > destinations and ports)

Wireshark - Destinations and Ports - Exercise.pcapng								
Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
Destinations and Ports	81420				0.0000	100%	14.9200	442267516.890
10.100.1.33	29387				0.0000	36.09%	1.1100	568415475.093
TCP	29387				0.0000	100.00%	1.1100	568415475.093
43514	15400				0.0000	52.40%	0.8300	568415468.711
48924	13987				0.0000	47.60%	0.8000	568415474.560
10.10.57.178	28984				0.0000	35.60%	1.2100	568415512.371
TCP	28939				0.0000	99.84%	1.2100	568415512.371

What is the max service request-response time of the DNS packets?

0.467897 (statistics > DNS > Service-stats > request response time)

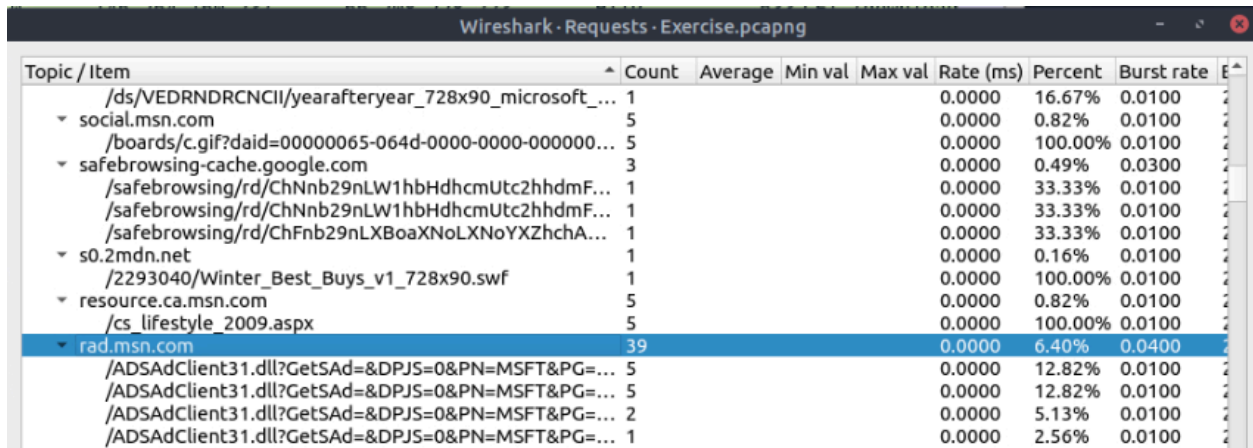


The screenshot shows the Wireshark Statistics window for the file 'DNS - Exercise.pcapng'. The 'Service Stats' section is expanded, showing a table of statistics for DNS service. The 'request-response time (secs)' row is highlighted in blue, showing a maximum value of 0.467897 ms.

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate	Burst start
A (Host Address)	102				0.0000	59.65%	0.0400	211538231.471
Class	171				0.0000	100.00%	0.1600	568415439.987
IN	171				0.0000	100.00%	0.1600	568415439.987
Service Stats	0				0.0000	100%	-	-
request-response time (secs)	85	0.07	0.000075	0.467897	0.0000		0.0800	568415439.988
no. of unsolicited responses	0				0.0000		-	-
no. of retransmissions	0				0.0000		-	-
Response Stats	0				0.0000	100%	-	-
no. of questions	170	1.00	1	1	0.0000		0.1600	568415439.988
no. of authorities	170	2.69	0	13	0.0000		0.1600	568415439.988
no. of answers	170	1.67	0	8	0.0000		0.1600	568415439.988

What is the number of HTTP Requests accomplished by "rad[.]msn[.]com"?

39 (statistics > http > requests)



The screenshot shows the Wireshark Statistics window for the file 'Exercise.pcapng', specifically the 'Requests' section. The 'rad.msn.com' entry is highlighted in blue, showing a count of 39 requests.

Topic / Item	Count	Average	Min val	Max val	Rate (ms)	Percent	Burst rate
/ds/VEDRNDRCNCII/yearafteryear_728x90_microsoft_...	1				0.0000	16.67%	0.0100
social.msn.com	5				0.0000	0.82%	0.0100
/boards/c.gif?daid=00000065-064d-0000-0000-000000...	5				0.0000	100.00%	0.0100
safebrowsing-cache.google.com	3				0.0000	0.49%	0.0300
/safebrowsing/rd/ChNnb29nLW1hbHdhcmUtc2hhdmF...	1				0.0000	33.33%	0.0100
/safebrowsing/rd/ChNnb29nLW1hbHdhcmUtc2hhdmF...	1				0.0000	33.33%	0.0100
/safebrowsing/rd/ChFnbn29nLXB0aXNoLXNoYXZhcA...	1				0.0000	33.33%	0.0100
s0.2mdn.net	1				0.0000	0.16%	0.0100
/2293040/Winter_Best_Buys_v1_728x90.swf	1				0.0000	100.00%	0.0100
resource.ca.msn.com	5				0.0000	0.82%	0.0100
/cs_lifestyle_2009.aspx	5				0.0000	100.00%	0.0100
rad.msn.com	39				0.0000	6.40%	0.0400
/ADSAdClient31.dll?GetSAd=&DPJS=0&PN=MSFT&PG=...	5				0.0000	12.82%	0.0100
/ADSAdClient31.dll?GetSAd=&DPJS=0&PN=MSFT&PG=...	5				0.0000	12.82%	0.0100
/ADSAdClient31.dll?GetSAd=&DPJS=0&PN=MSFT&PG=...	2				0.0000	5.13%	0.0100
/ADSAdClient31.dll?GetSAd=&DPJS=0&PN=MSFT&PG=...	1				0.0000	2.56%	0.0100

Task 4 - Packet Filtering | Principles

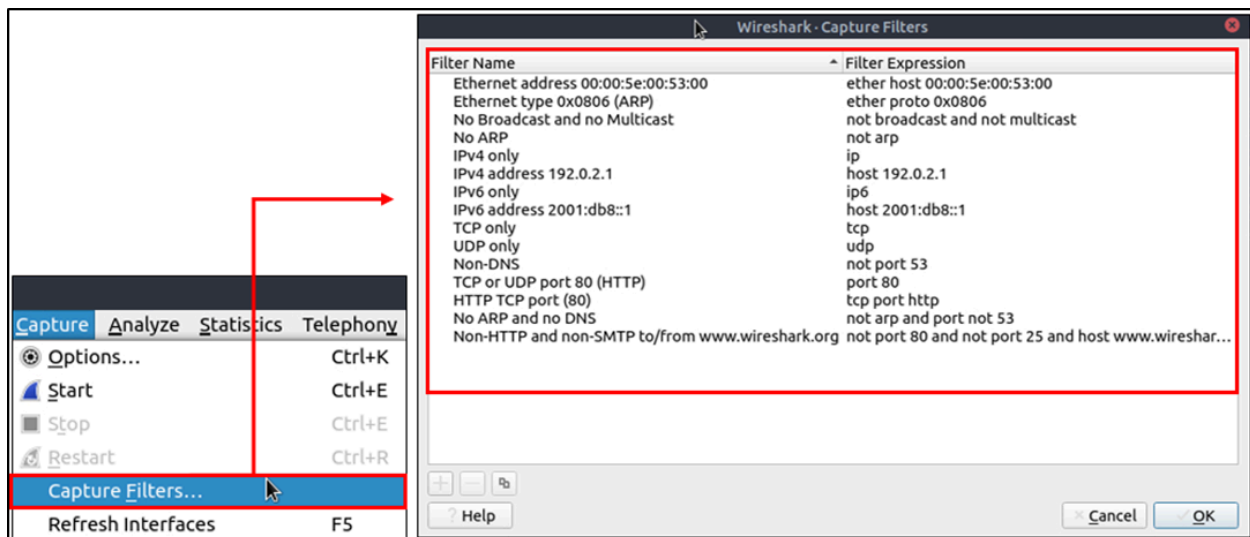
Packet Filtering: There are two types of filters in Wireshark. While both use similar syntax, they are used for different purposes.

Capture Filters	This type of filter is used to save only a specific part of the traffic. It is set before capturing traffic and not changeable during the capture.
Display Filters	This type of filter is used to investigate packets by reducing the number of visible packets, and it is changeable during the capture

Capture Filter Syntax

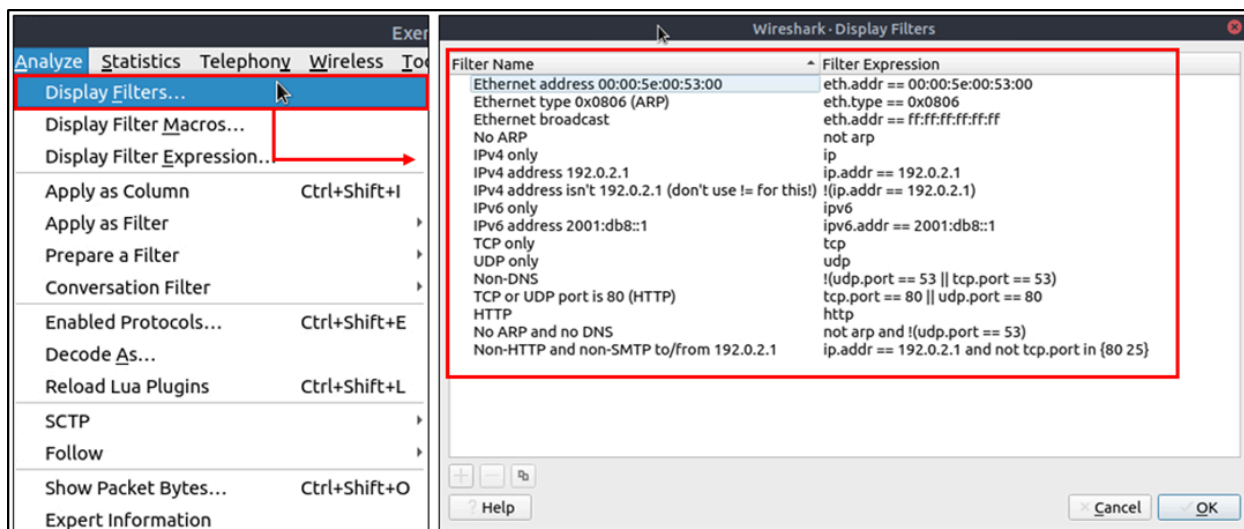
- **Scope:** host, net, port and portrange.
- **Direction:** src, dst, src or dst, src and dst,
- **Protocol:** ether, wlan, ip, ip6, arp, rarp, tcp and udp.
- **Sample filter to capture port 80 traffic:** `tcp port 80`

A quick reference is available under the "**Capture → Capture Filters**" menu.



Display Filter Syntax

- **Sample filter to capture port 80 traffic:** `tcp.port == 80` "**Analyse → Display Filters**"



Comparison Operators

English	C-Like	Description	Example
eq	==	Equal	<code>ip.src == 10.10.10.100</code>
ne	!=	Not equal	<code>ip.src != 10.10.10.100</code>
gt	>	Greater than	<code>ip.ttl > 250</code>
lt	<	Less Than	<code>ip.ttl < 10</code>
ge	>=	Greater than or equal to	<code>ip.ttl >= 0xFA</code>
le	<=	Less than or equal to	<code>ip.ttl <= 0xA</code>

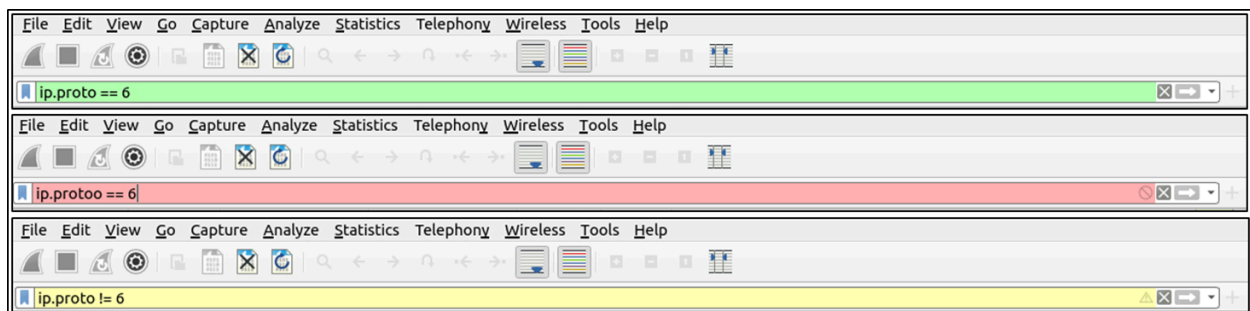
Logical Expressions

English	C-Like	Description	Example
and	&&	Logical AND	<code>(ip.src == 10.10.10.100) AND (ip.src == 10.10.10.111)</code>
or		Logical OR	<code>(ip.src == 10.10.10.100) OR (ip.src == 10.10.10.111)</code>
not	!	Logical NOT	<code>!(ip.src == 10.10.10.222)</code> Note: Usage of <code>!=value</code> is deprecated; using it could provide inconsistent results. Using the <code>!(value)</code> style is suggested for more consistent results.

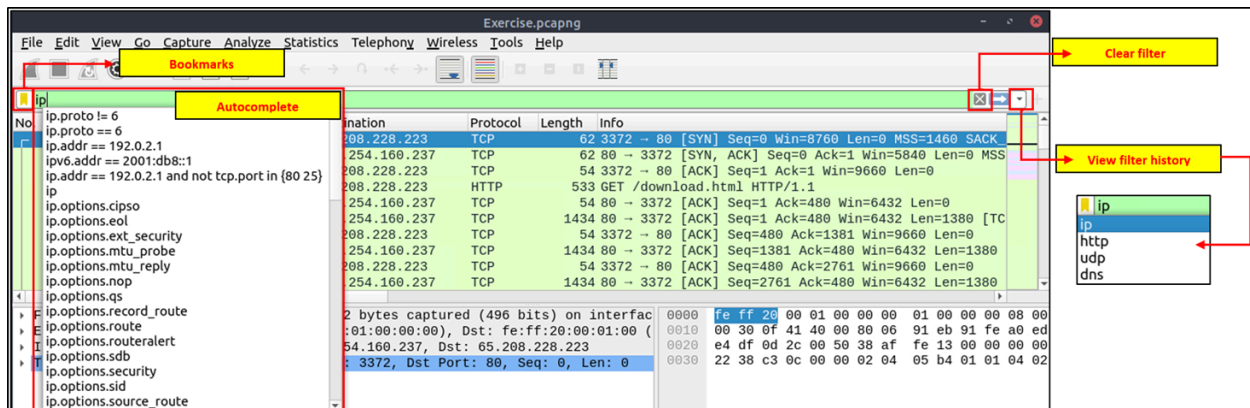
Packet Filter Toolbar

- Packet filters are defined in lowercase.
- Packet filters have an autocomplete feature to break down protocol details, and each detail is represented by a "dot".
- Packet filters have a three-colour representation explained below.

Green	Valid filter
Red	Invalid filter
Yellow	Warning filter. This filter works, but it is unreliable, and it is suggested to change it with a valid filter.



Filter toolbar features are shown below.



Task 5 - Packet Filtering | Protocol Filters

Protocol Filters: Wireshark supports 3000 protocols and allows packet-level investigation by filtering the protocol fields. This task shows the creation and usage of filters against different protocol fields.

IP Filters: IP filters help analysts filter the traffic according to the IP level information from the packets (Network layer of the OSI model). This is one of the most commonly used filters in Wireshark. These filters filter network-level information like IP addresses, version, time to live, type of service, flags, and checksum values.

Filter	Description
<code>ip</code>	Show all IP packets.
<code>ip.addr == 10.10.10.111</code>	Show all packets containing IP address 10.10.10.111.
<code>ip.addr == 10.10.10.0/24</code>	Show all packets containing IP addresses from 10.10.10.0/24 subnet.
<code>ip.src == 10.10.10.111</code>	Show all packets originated from 10.10.10.111
<code>ip.dst == 10.10.10.111</code>	Show all packets sent to 10.10.10.111
ip.addr vs ip.src/ip.dst	Note: The ip.addr filters the traffic without considering the packet direction. The ip.src/ip.dst filters the packet depending on the packet direction.

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with icons for Start, Stop, Restart, Options, Open, Save, Close, Reload, Find Packet..., Previous Packet, Next Packet, Go to Packet..., First Packet, Last Packet, Auto Scroll in Live Capture, Colorize Packet List, and Zoom In. The main window is divided into three panes: Packet List, Packet Details, and Packet Bytes.

Packet List: Shows a list of 21 captured packets. The columns are No., Time, Source, Destination, Protocol, Length, Type, and Info. Packet 1 is selected, showing details of an Ethernet II frame, an IPv4 packet, and a TCP segment.

Packet Details: Shows the hierarchical structure of the selected packet. The selected packet is an Ethernet II frame, Src: Xerox_00:00:00:00 (00:00:01:00:00:00), Dst: fe:ff:20:00:01:00. The details pane shows the Ethernet II frame, Internet Protocol Version 4, and Transmission Control Protocol.

Packet Bytes: Shows the raw data of the selected packet in hexadecimal and ASCII. The data is displayed in a hex dump format, with the ASCII representation shown on the right.

TCP and UDP Filters: Enable analysts to filter transport-layer traffic based on details such as ports, sequence numbers, flags, timestamps, packet length, and protocol errors to investigate specific network communications.

Filter	Description	Filter	Expression
<code>tcp.port == 80</code>	Show all TCP packets with port 80	<code>udp.port == 53</code>	Show all UDP packets with port 53
<code>tcp.srcport == 1234</code>	Show all TCP packets originating from port 1234	<code>udp.srcport == 1234</code>	Show all UDP packets originating from port 1234
<code>tcp.dstport == 80</code>	Show all TCP packets sent to port 80	<code>udp.dstport == 5353</code>	Show all UDP packets sent to port 5353

The image shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with icons for Start, Stop, Restart, Options, Open, Save, Close, Reload, Find Packet..., Previous Packet, Next Packet, Go to Packet..., First Packet, Last Packet, Auto Scroll in Live Capture, Colorize Packet List, and Zoom In. The main packet list pane shows a table of captured packets with columns for No., Time, Source, Destination, Protocol, Length, Type, and Info. The selected packet (No. 1) is a TCP packet from 145.254.160.237 to 65.208.228.223 on port 80. The packet details pane on the right shows the structure of the selected packet: Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII.

Application Level Protocol Filters | HTTP and DNS

Filter	Description	Filter	Description
--------	-------------	--------	-------------

<code>http</code>	Show all HTTP packets	<code>dns</code>	Show all DNS packets
<code>http.response.code == 200</code>	Show all packets with HTTP response code "200"	<code>dns.flags.response == 0</code>	Show all DNS requests
<code>http.request.method == "GET"</code>	Show all HTTP GET requests	<code>dns.flags.response == 1</code>	Show all DNS responses
<code>http.request.method == "POST"</code>	Show all HTTP POST requests	<code>dns.qry.type == 1</code>	Show all DNS "A" records

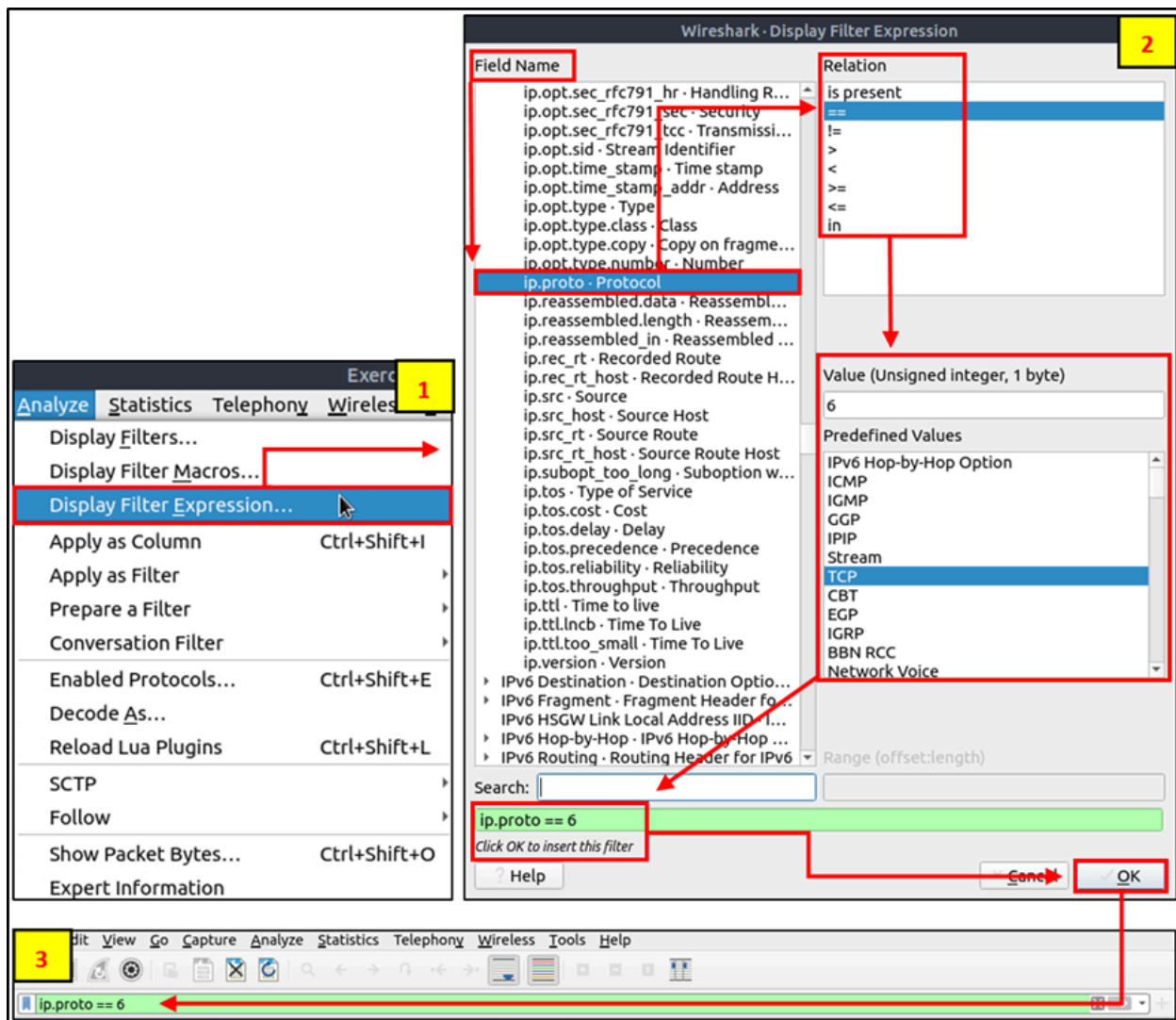
The image shows the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with icons for Start, Stop, Restart, Options, Open, Save, Close, Reload, Find Packet..., Previous Packet, Next Packet, Go to Packet..., First Packet, Last Packet, Auto Scroll in Live Capture, Colorize Packet List, and Zoom In. A display filter bar shows 'Apply a display filter ... <Ctrl-/>'. The packet list pane shows a table of captured packets with columns for No., Time, Source, Destination, Protocol, Length, Type, and Info. Packet 13 is selected, showing a DNS Standard query. The packet details pane on the right shows the hierarchy: Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Domain Name System (query). The packet bytes pane at the bottom shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Type	Info
3	2004-05-13 10:17:08.222534	145.254.160.237	65.208.228.223	TCP	54	IPv4	3372 → 80 [ACK] Seq=1 Ac
4	2004-05-13 10:17:08.222534	145.254.160.237	65.208.228.223	HTTP	533	IPv4	GET /download.html HTTP/
5	2004-05-13 10:17:08.783340	65.208.228.223	145.254.160.237	TCP	54	IPv4	80 → 3372 [ACK] Seq=1 Ac
6	2004-05-13 10:17:08.993643	65.208.228.223	145.254.160.237	TCP	1434	IPv4	80 → 3372 [ACK] Seq=1 Ac
7	2004-05-13 10:17:09.123830	145.254.160.237	65.208.228.223	TCP	54	IPv4	3372 → 80 [ACK] Seq=480
8	2004-05-13 10:17:09.123830	65.208.228.223	145.254.160.237	TCP	1434	IPv4	80 → 3372 [ACK] Seq=1381
9	2004-05-13 10:17:09.324118	145.254.160.237	65.208.228.223	TCP	54	IPv4	3372 → 80 [ACK] Seq=480
10	2004-05-13 10:17:09.754737	65.208.228.223	145.254.160.237	TCP	1434	IPv4	80 → 3372 [ACK] Seq=2761
11	2004-05-13 10:17:09.864896	65.208.228.223	145.254.160.237	TCP	1434	IPv4	80 → 3372 [PSH, ACK] Seq
12	2004-05-13 10:17:09.864896	145.254.160.237	65.208.228.223	TCP	54	IPv4	3372 → 80 [ACK] Seq=480
13	2004-05-13 10:17:09.864896	145.254.160.237	145.253.2.203	DNS	89	IPv4	Standard query 0x0023 A
14	2004-05-13 10:17:09.945011	65.208.228.223	145.254.160.237	TCP	1434	IPv4	80 → 3372 [ACK] Seq=5521
15	2004-05-13 10:17:10.125270	145.254.160.237	65.208.228.223	TCP	54	IPv4	3372 → 80 [ACK] Seq=480
16	2004-05-13 10:17:10.205385	65.208.228.223	145.254.160.237	TCP	1434	IPv4	80 → 3372 [ACK] Seq=6901
17	2004-05-13 10:17:10.225414	145.253.2.203	145.254.160.237	DNS	188	IPv4	Standard query response
18	2004-05-13 10:17:10.295515	145.254.160.237	216.239.59.99	HTTP	775	IPv4	GET /pagead/ads?client=c
19	2004-05-13 10:17:10.325558	145.254.160.237	65.208.228.223	TCP	54	IPv4	3372 → 80 [ACK] Seq=480
20	2004-05-13 10:17:10.686076	65.208.228.223	145.254.160.237	TCP	1434	IPv4	80 → 3372 [ACK] Seq=8281
21	2004-05-13 10:17:10.806249	65.208.228.223	145.254.160.237	TCP	1434	IPv4	80 → 3372 [PSH, ACK] Seq
22	2004-05-13 10:17:10.806249	145.254.160.237	65.208.228.223	TCP	54	IPv4	3372 → 80 [ACK] Seq=480
23	2004-05-13 10:17:10.946451	65.208.228.223	145.254.160.237	TCP	1434	IPv4	80 → 3372 [ACK] Seq=1104

Frame 13: 89 bytes on wire (712 bits), 89 bytes captured (712 bits) on interface 0
 Ethernet II, Src: Xerox_00:00:00:00:00:00, Dst: fe:ff:20:00:00:00
 Internet Protocol Version 4, Src: 145.254.160.237, Dst: 145.253.2.203
 User Datagram Protocol, Src Port: 3009, Dst Port: 53
 Domain Name System (query)

0000 fe ff 20 00 01 00 00 00 01 00 00 00 08 00 45 00
 0010 00 4b 0f 49 00 00 80 11 63 a5 91 fe a0 ed 91 fd
 0020 02 cb 0b c1 00 35 00 37 10 af 00 23 01 00 00 01
 0030 00 00 00 00 00 00 07 70 61 67 65 61 64 32 11 67
 0040 6f 6f 67 6c 65 73 79 6e 64 69 63 61 74 69 6f 6e
 0050 03 63 6f 6d 00 01 00 01

Display Filter Expressions: Provides a built-in filter builder that helps analysts create Wireshark display filters by showing available protocol fields, value types, and predefined options, making complex packet analysis easier. **"Analyze → Display Filter Expression"**



"View → Coloring Rules" menu to assign colours to highlight your display filter results.

Answer the questions below

What is the number of IP packets? 81420

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	145.254.160.237	65.208.228.223	TCP	62	3372 → 80 [SYN]
2	0.911310	65.208.228.223	145.254.160.237	TCP	62	80 → 3372 [SYN]
3	0.911310	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK]
4	0.911310	145.254.160.237	65.208.228.223	HTTP	533	GET /download
5	1.472116	65.208.228.223	145.254.160.237	TCP	54	80 → 3372 [ACK]
6	1.682419	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK]
7	1.812606	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK]
8	1.812606	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK]
9	2.012894	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK]
10	2.443513	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK]
11	2.553672	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [PSH]
12	2.553672	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK]
13	2.553672	145.254.160.237	145.253.2.203	DNS	89	Standard query
14	2.633787	65.208.228.223	145.254.160.237	TCP	1434	80 → 3372 [ACK]
15	2.814046	145.254.160.237	65.208.228.223	TCP	54	3372 → 80 [ACK]

Frame 1: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface 0	0000	fe ff 20 00 01 00 00 00	01 00 00 00 00 00 45 00
Ethernet II, Src: Xerox_00:00:00 (00:00:01:00:00:00), Dst: 02:00:00:00:00:00	0010	00 30 0f 41 40 00 80 06	91 eb 91 fe a0 ed 41 d0
Internet Protocol Version 4, Src: 145.254.160.237, Dst: 65.208.228.223	0020	e4 df 0d 2c 00 50 38 af	fe 13 00 00 00 00 70 00
Transmission Control Protocol, Src Port: 3372, Dst Port: 80	0030	22 38 c3 0c 00 00 02 04	05 b4 01 01 04 02

What is the number of packets with a "TTL value less than 10"?

66(Analyse > display filter expression)

ip.ttl < 10

No.	Time	Source	Destination	Protocol	Length	Info
423	211538132.29...	192.168.3.131	239.255.255.250	SSDP	167	M-SEARCH * HT
424	211538132.29...	192.168.3.131	239.255.255.250	SSDP	165	M-SEARCH * HT
727	211538135.29...	192.168.3.131	239.255.255.250	SSDP	167	M-SEARCH * HT
728	211538135.29...	192.168.3.131	239.255.255.250	SSDP	165	M-SEARCH * HT
815	211538138.29...	192.168.3.131	239.255.255.250	SSDP	167	M-SEARCH * HT
816	211538138.29...	192.168.3.131	239.255.255.250	SSDP	165	M-SEARCH * HT
1932	211538145.75...	192.168.3.131	224.0.0.252	LLMNR	64	Standard quer
1941	211538145.85...	192.168.3.131	224.0.0.252	LLMNR	64	Standard quer
4025	211538213.02...	172.16.255.1	224.0.0.252	LLMNR	64	Standard quer
4036	211538213.12...	172.16.255.1	224.0.0.252	LLMNR	64	Standard quer
4143	211538217.02...	172.16.255.1	239.255.255.250	SSDP	175	M-SEARCH * HT
4144	211538217.02...	172.16.255.1	239.255.255.250	SSDP	174	M-SEARCH * HT
5998	211538231.84...	10.0.2.15	10.0.2.2	SSDP	174	M-SEARCH * HT
5999	211538231.84...	10.0.2.15	10.0.2.2	SSDP	175	M-SEARCH * HT
6003	211538231.85...	10.0.2.2	10.0.2.15	ICMP	202	Time-to-live

▶ Frame 423: 167 bytes on wire (1336 bits), 16
 ▶ Ethernet II, Src: Micro-St_9a:f1:f5 (40:61:8
 ▶ Internet Protocol Version 4, Src: 192.168.3.
 ▶ User Datagram Protocol, Src Port: 57757, Dst
 ▶ Simple Service Discovery Protocol

```

0000  01 00 5e 7f ff fa 40 61 86 9a f1 f5 08 00 45 0
0010  00 99 68 f9 00 00 01 11 9c 35 c0 a8 03 83 ef
0020  ff fa e1 9d 07 6c 00 85 5e 7f 4d 2d 53 45 41
0030  43 48 20 2a 20 48 54 54 50 2f 31 2e 31 0d 0a
0040  6f 73 74 3a 32 33 39 2e 32 35 35 2e 32 35 35
0050  32 35 30 3a 31 39 30 30 0d 0a 53 54 3a 75 72
0060  3a 73 63 68 65 6d 61 73 2d 75 70 6e 70 2d 6f
0070  67 3a 64 65 76 69 63 65 3a 4d 65 64 69 61 52
0080  6e 64 65 72 65 72 3a 31 0d 0a 4d 61 6e 3a 22
0090  73 64 70 3a 64 69 73 63 6f 76 65 72 22 0d 0a
00a0  58 3a 33 0d 0a 0d 0a
  
```

Exercise.pcapng Packets: 81530 - Displayed: 66 (0.1%) - Comments: 1 Profile: Default

What is the number of packets which uses "TCP port 4444"? 632

tcp.port == 4444						
No.	Time	Source	Destination	Protocol	Length	Info
14345	442267516.82...	192.168.1.100	104.131.15.86	TCP	66	60745 → 4444
14389	442267516.83...	192.168.1.100	139.99.8.158	TCP	66	60788 → 4444
14401	442267516.83...	192.168.1.100	139.99.9.157	TCP	66	60800 → 4444
14406	442267516.83...	192.168.1.100	139.99.8.222	TCP	66	60805 → 4444
14413	442267516.83...	192.168.1.100	139.99.9.175	TCP	66	60812 → 4444
14443	442267516.83...	192.168.1.100	149.56.26.221	TCP	66	60842 → 4444
14447	442267516.83...	192.168.1.100	158.69.246.183	TCP	66	60846 → 4444
14456	442267516.84...	192.168.1.100	158.69.25.62	TCP	66	60855 → 4444
14462	442267516.84...	192.168.1.100	158.69.25.71	TCP	66	60862 → 4444
14468	442267516.84...	192.168.1.100	158.69.25.77	TCP	66	60868 → 4444
14573	442267516.85...	192.168.1.100	167.114.210.141	TCP	66	60952 → 4444
14584	442267516.85...	192.168.1.100	176.31.117.82	TCP	66	60964 → 4444
14591	442267516.85...	192.168.1.100	165.227.194.221	TCP	66	60971 → 4444
14637	442267516.86...	192.168.1.100	178.62.205.21	TCP	66	61000 → 4444
14647	442267516.86...	192.168.1.100	188.165.199.78	TCP	66	61011 → 4444

Frame 14345: 66 bytes on wire (528 bits), 66	0000	d0 60 8c 2e f0 5e 68 f7 28 ce 28 5f 08 00 45 0
Ethernet II, Src: LCFCHeFe_ce:28:5f (68:f7:2	0010	00 34 0a d5 40 00 80 06 00 00 c0 a8 01 64 68 8
Internet Protocol Version 4, Src: 192.168.1.	0020	0f 56 ed 49 11 5c 08 20 e2 14 00 00 00 80 0
Transmission Control Protocol, Src Port: 607	0030	fa f0 3a 0c 00 00 02 04 05 b4 01 03 03 08 01 0
	0040	04 02

What is the number of "HTTP GET" requests sent to port "80"? 527

Filter: http.request.method == GET and tcp.dstport == 80

No.	Time	Source	Destination	Protocol	Length	Info
11449	211538347.64...	192.168.3.131	208.82.236.129	HTTP	684	GET /search/m
11536	211538348.59...	192.168.3.131	72.14.213.138	HTTP	1437	GET /safebrow
11539	211538348.61...	192.168.3.131	72.14.213.138	HTTP	1431	GET /safebrow
11541	211538348.64...	192.168.3.131	72.14.213.138	HTTP	1429	GET /safebrow
11707	211538352.20...	192.168.3.131	208.82.236.129	HTTP	684	GET /search/m
11797	211538353.21...	192.168.3.131	174.36.30.111	HTTP	242	GET /subscrib
11831	211538355.13...	192.168.3.131	208.82.236.129	HTTP	650	GET /bnc/mcy/
11842	211538355.21...	192.168.3.131	208.82.236.130	HTTP	497	GET /3k83pb3l
11858	211538355.35...	192.168.3.131	209.17.73.30	HTTP	487	GET /albums/y
11861	211538355.35...	192.168.3.131	209.17.73.30	HTTP	487	GET /albums/y
12278	211538356.48...	192.168.3.131	208.82.236.129	HTTP	650	GET /van/mcy/
12287	211538356.56...	192.168.3.131	208.82.236.130	HTTP	497	GET /3m13p13l
12298	211538356.62...	192.168.3.131	208.82.236.130	HTTP	497	GET /3kd3m13o
13111	211538374.27...	192.168.3.131	208.82.236.129	HTTP	650	GET /van/mcy/
13757	211538400.86...	172.16.255.1	204.9.163.158	HTTP	119	GET / HTTP/1.

Frame 13757: 119 bytes on wire (952 bits), 1 Ethernet II, Src: QuantaCo_51:4f:a9 (00:1e:61:49:a3:9e), Dst: 08:00:2b:01:02:00, Internet Protocol Version 4, Src: 172.16.255.1, Destination: 204.9.163.158, Transmission Control Protocol, Src Port: 10700, Hypertext Transfer Protocol

```

0000  00 1a 8c 10 ad 30 00 1e 68 51 4f a9 08 00 45 00
0010  00 69 7e 8d 40 00 80 06 61 47 ac 10 ff 01 cc 00
0020  a3 9e 29 da 00 50 79 c2 d2 06 27 4d a0 dc 50 00
0030  40 29 6d 15 00 00 47 45 54 20 2f 20 48 54 54 54
0040  2f 31 2e 30 0d 0a 48 6f 73 74 3a 20 63 6f 6e 6e
0050  2e 73 6b 79 70 65 2e 63 6f 6d 0d 0a 55 73 65 73
0060  2d 41 67 65 6e 74 3a 20 53 6b 79 70 65 20 57 20
0070  53 50 72 0d 0a 0d 0a

```

Exercise.pcapng Packets: 81530 · Displayed: 527 (0.6%) · Comments: 1 Profile: Default

What is the number of type A DNS Queries? 51

dns.qry.type == 1 and dns.flags.response == 1

No.	Time	Source	Destination	Protocol	Length	Info
11001	211538338.62...	10.0.2.3	10.0.2.15	DNS	281	Standard quer
11003	211538338.64...	10.0.2.3	10.0.2.15	DNS	434	Standard quer
11141	211538341.47...	10.0.2.3	10.0.2.15	DNS	400	Standard quer
11145	211538341.54...	10.0.2.3	10.0.2.15	DNS	397	Standard quer
11247	211538342.26...	10.0.2.3	10.0.2.15	DNS	227	Standard quer
11307	211538342.68...	10.0.2.3	10.0.2.15	DNS	345	Standard quer
11329	211538342.84...	10.0.2.3	10.0.2.15	DNS	289	Standard quer
11374	211538346.33...	10.0.2.3	10.0.2.15	DNS	241	Standard quer
11402	211538346.96...	10.0.2.3	10.0.2.15	DNS	160	Standard quer
11440	211538347.54...	10.0.2.3	10.0.2.15	DNS	361	Standard quer
11560	211538349.05...	10.0.2.3	10.0.2.15	DNS	359	Standard quer
11610	211538350.97...	10.0.2.3	10.0.2.15	DNS	213	Standard quer
12760	211538368.22...	10.0.2.3	10.0.2.15	DNS	196	Standard quer
13233	211538379.55...	10.0.2.3	10.0.2.15	DNS	195	Standard quer
13540	211538394.53...	10.0.2.3	10.0.2.15	DNS	195	Standard quer

Frame 13540: 195 bytes on wire (1560 bits),
 Ethernet II, Src: RealtekU_12:35:02 (52:54:00:12:35:02), Dst: 08:00:27:cc:3f:1b (08:00:27:cc:3f:1b)
 Internet Protocol Version 4, Src: 10.0.2.3, Dst: 10.0.2.15
 User Datagram Protocol, Src Port: 53, Dst Port: 53
 Domain Name System (response)

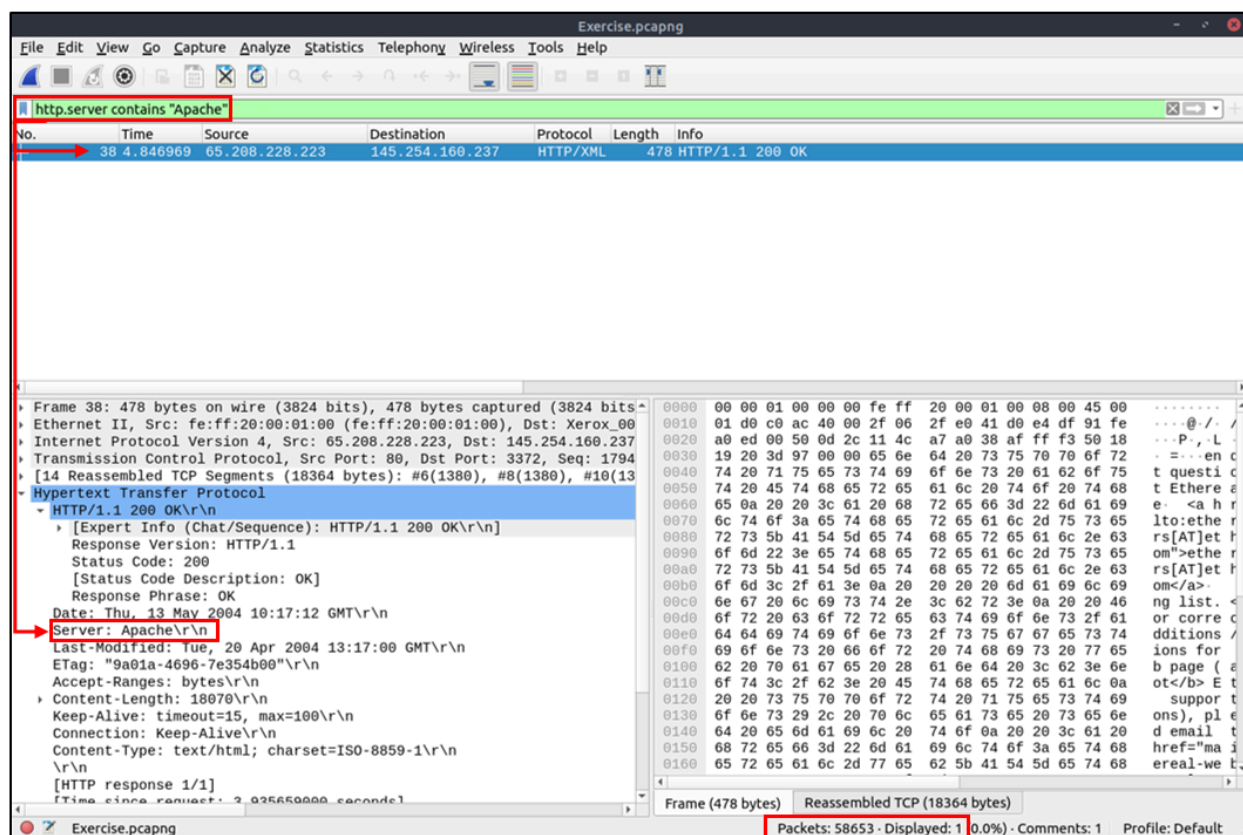
0000 08 00 27 cc 3f 1b 52 54 00 12 35 02 08 00 45 00
 0010 00 b5 00 64 00 00 40 11 61 c3 0a 00 02 03 0a 00
 0020 02 0f 00 35 e7 e5 00 a1 e6 78 1e ef 81 80 00 00
 0030 00 03 00 02 00 00 04 69 6d 67 34 07 63 61 74 66
 0040 6c 6f 67 05 76 69 64 65 6f 03 6d 73 6e 03 63 66
 0050 6d 00 00 01 00 01 c0 0c 00 05 00 01 00 00 03 80
 0060 00 19 09 6d 73 6e 76 69 64 77 65 62 02 76 6f 00
 0070 6d 73 65 63 6e 64 03 6e 65 74 00 c0 38 00 01 00
 0080 01 00 00 01 12 00 04 46 25 81 22 c0 38 00 01 00
 0090 01 00 00 01 12 00 04 46 25 81 20 c0 45 00 02 00
 00a0 01 00 00 01 12 00 08 05 64 6e 73 31 34 c0 45 c0
 00b0 45 00 02 00 01 00 00 01 12 00 08 05 64 6e 73 31
 00c0 31 c0 45

Exercise.pcapng Packets: 81530 · Displayed: 51 (0.1%) · Comments: 1 Profile: Default

Task 6 - Advanced Filtering

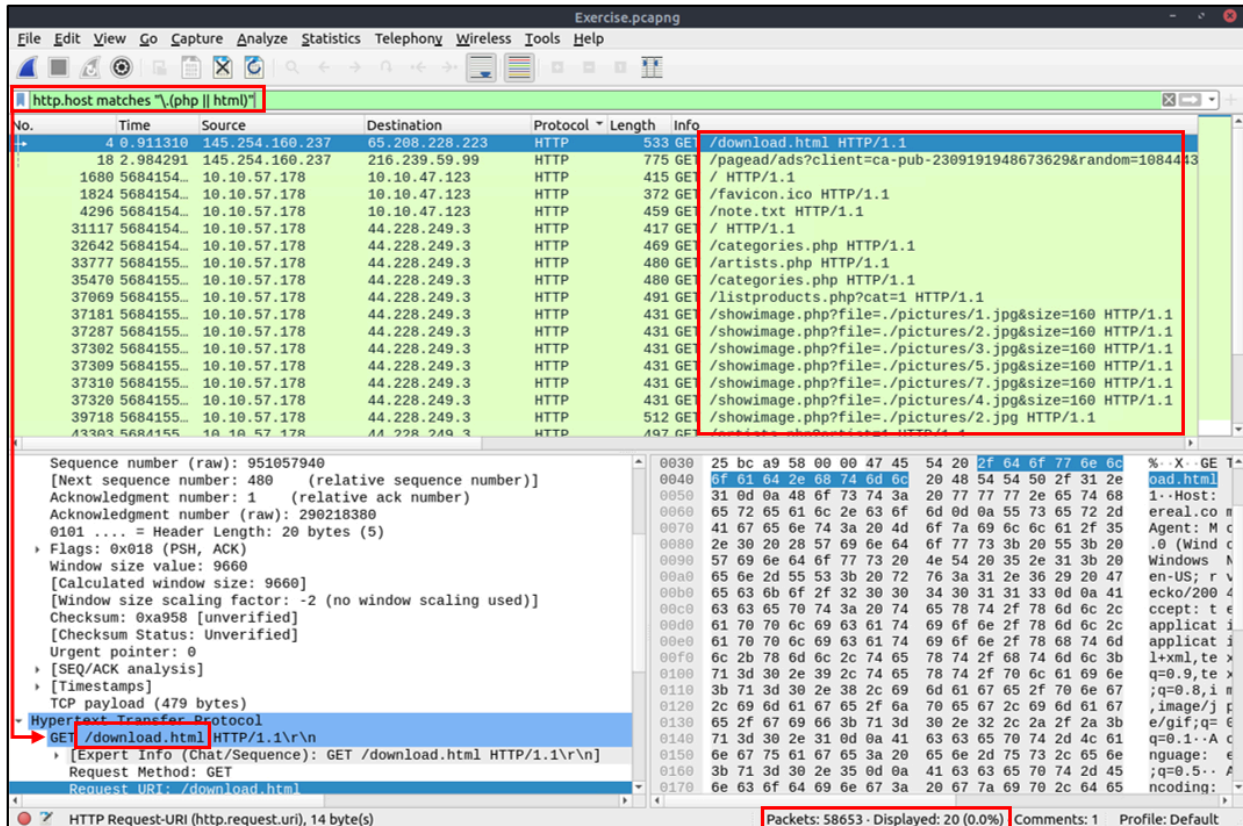
Filter: "contains"

Filter	contains
Type	Comparison Operator
Description	Search a value inside packets. It is case-sensitive and provides similar functionality to the "Find" option by focusing on a specific field.
Example	Find all "Apache" servers.
Workflow	List all HTTP packets where packets' "server" field contains the "Apache" keyword.
Usage	<code>http.server contains "Apache"</code>



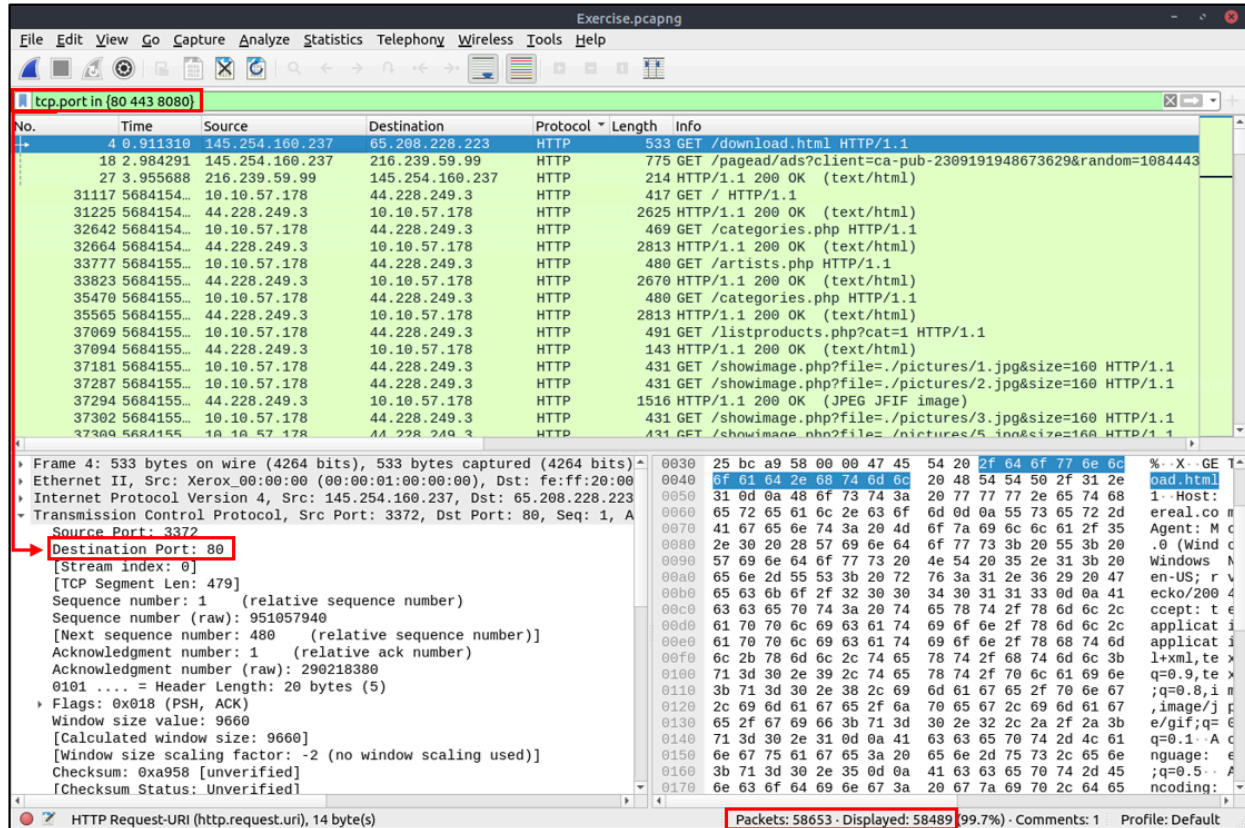
Filter: "matches"

Filter	matches
Type	Comparison Operator
Description	Search a pattern of a regular expression. It is case insensitive, and complex queries have a margin of error.
Example	Find all .php and .html pages.
Workflow	List all HTTP packets where packets' "host" fields match keywords ".php" or ".html".
Usage	<code>http.host matches "\.(php html)"</code>



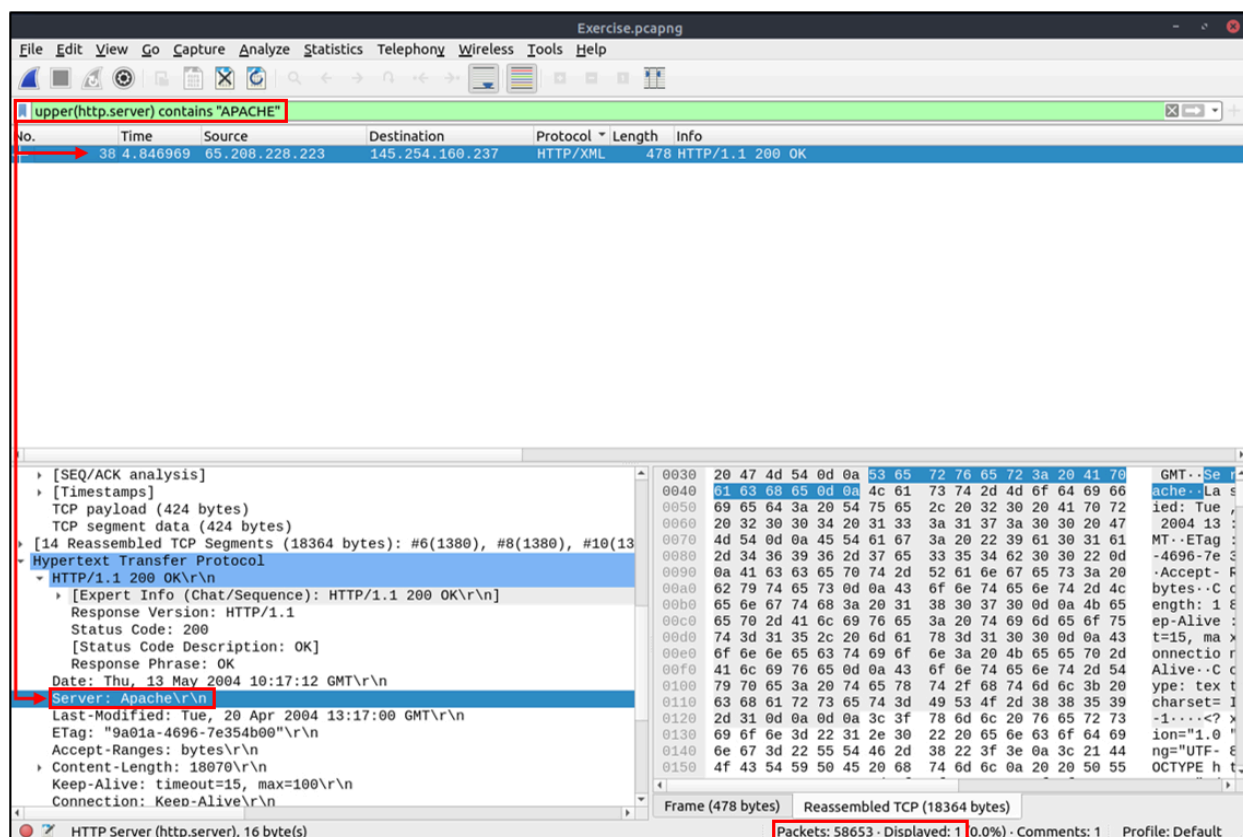
Filter: "in"

Filter	in
Type	Set Membership
Description	Search a value or field inside of a specific scope/range.
Example	Find all packets that use ports 80, 443 or 8080.
Workflow	List all TCP packets where packets' "port" fields have values 80, 443 or 8080.
Usage	<code>tcp.port in {80 443 8080}</code>



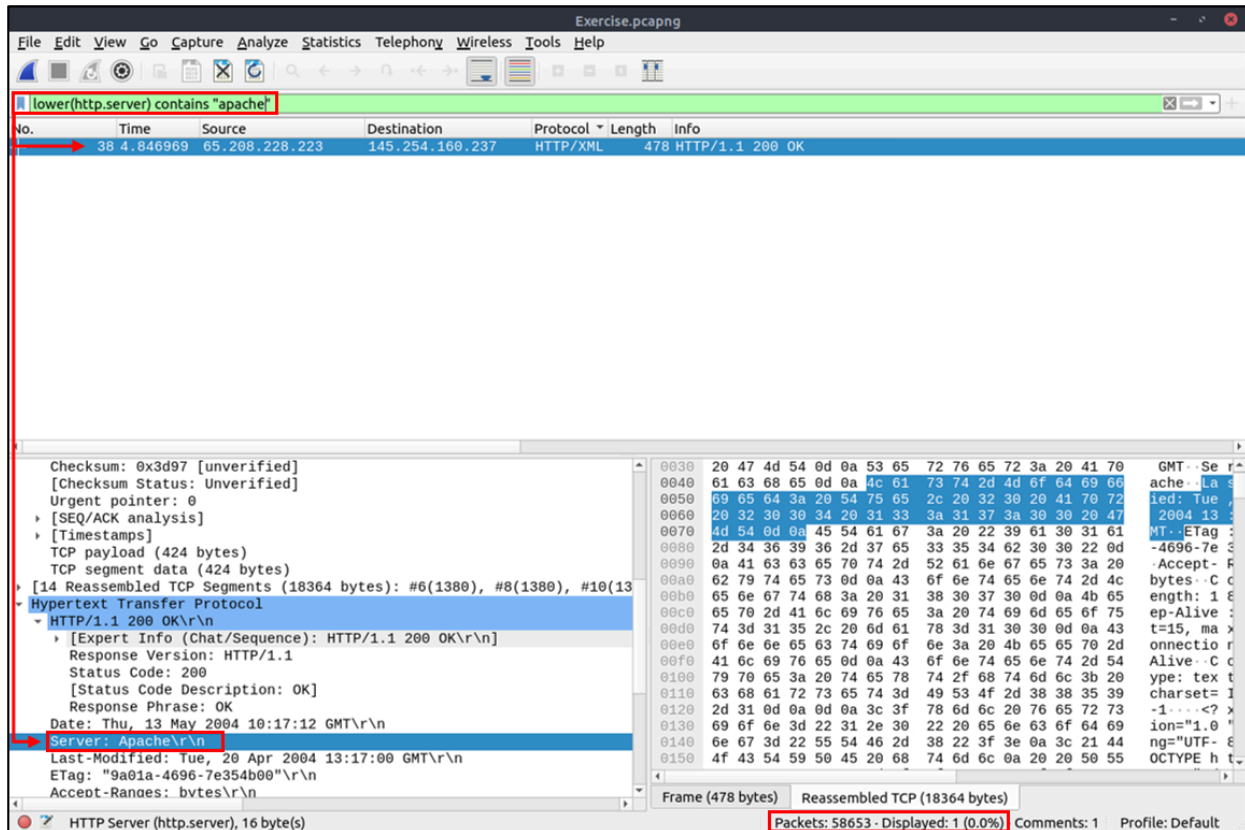
Filter: "upper"

Filter	upper
Type	Function
Description	Convert a string value to uppercase.
Example	Find all "APACHE" servers.
Workflow	Convert all HTTP packets' "server" fields to uppercase and list packets that contain the "APACHE" keyword.
Usage	<code>upper(http.server) contains "APACHE"</code>



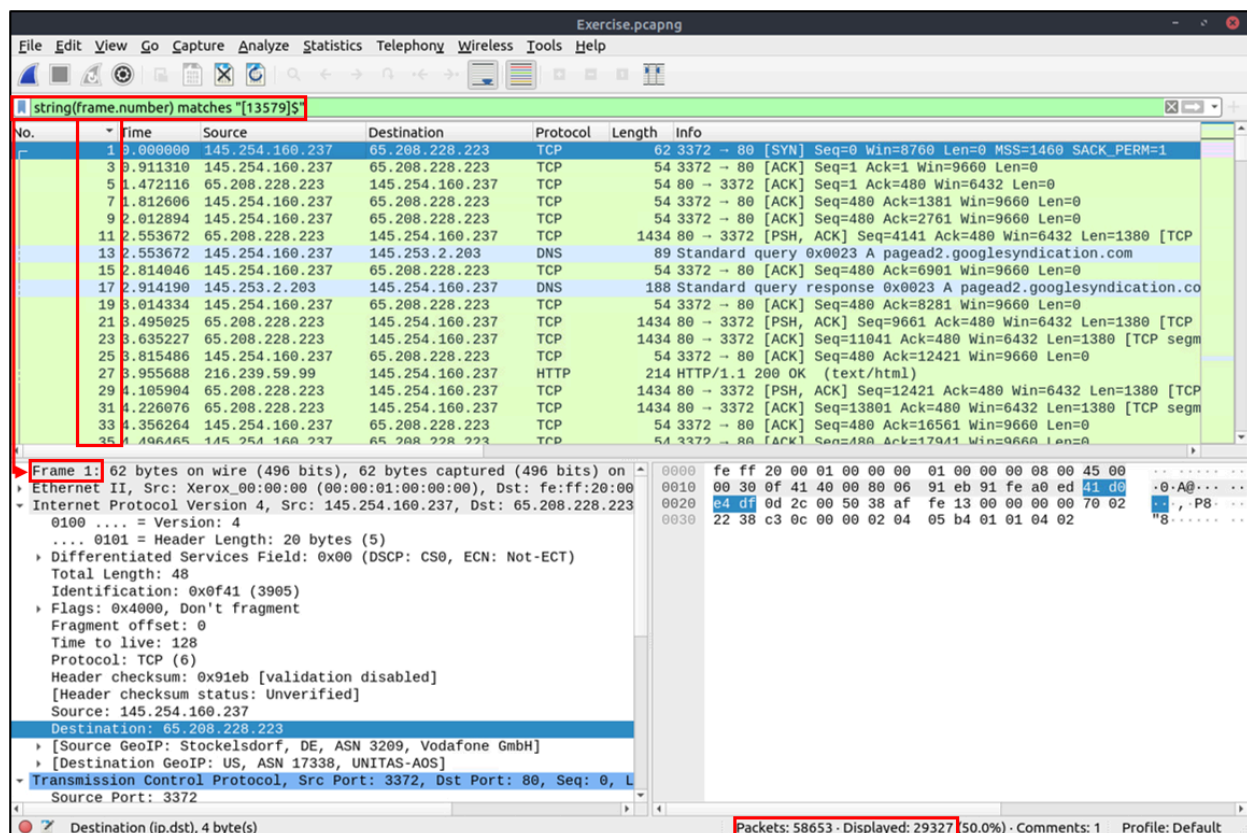
Filter: "lower"

Filter	lower
Type	Function
Description	Convert a string value to lowercase.
Example	Find all "apache" servers.
Workflow	Convert all HTTP packets' "server" fields info to lowercase and list packets that contain the "apache" keyword.
Usage	<code>lower(http.server) contains "apache"</code>

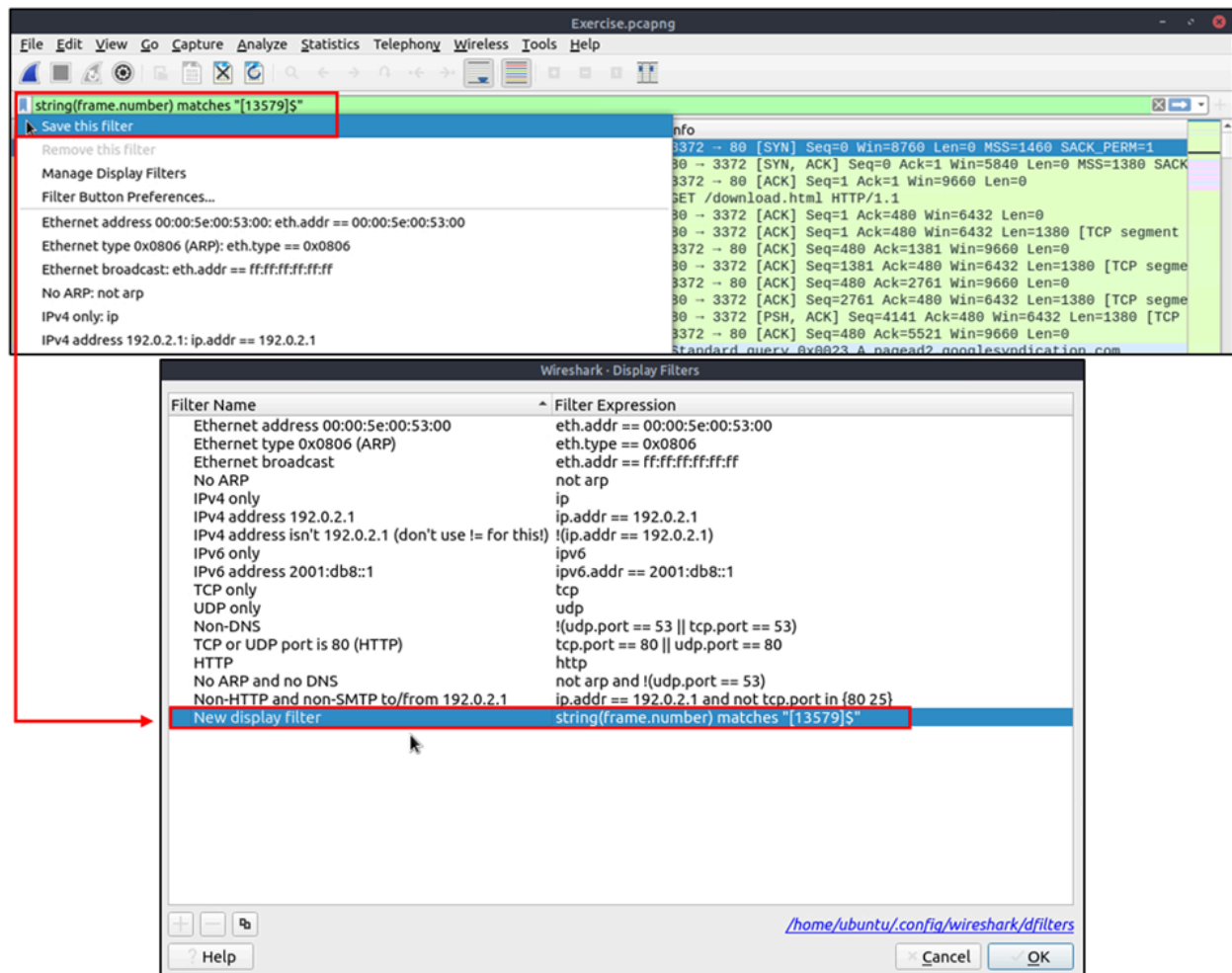


Filter: "string"

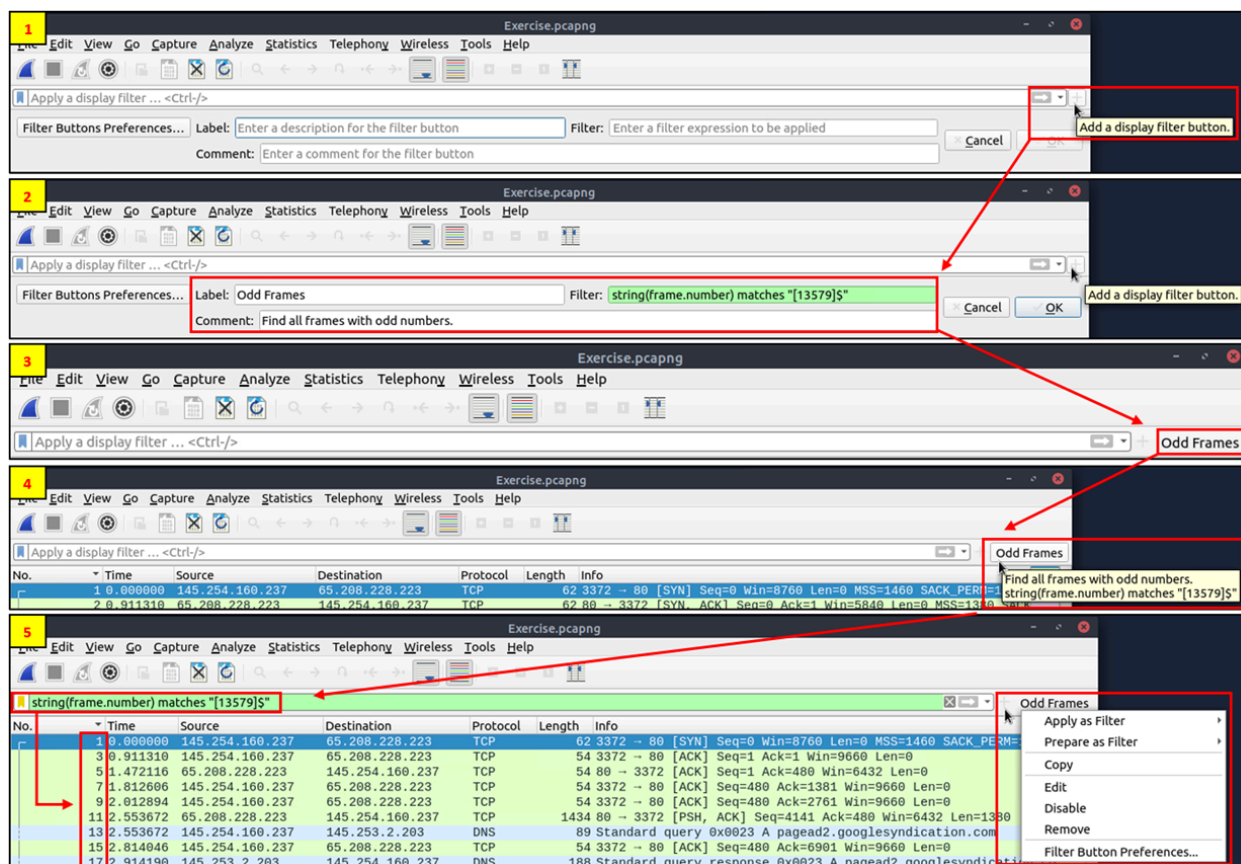
Filter	string
Type	Function
Description	Convert a non-string value to a string.
Example	Find all frames with odd numbers.
Workflow	Convert all "frame number" fields to string values, and list frames end with odd values.
Usage	<code>string(frame.number) matches "[13579]\$"</code>



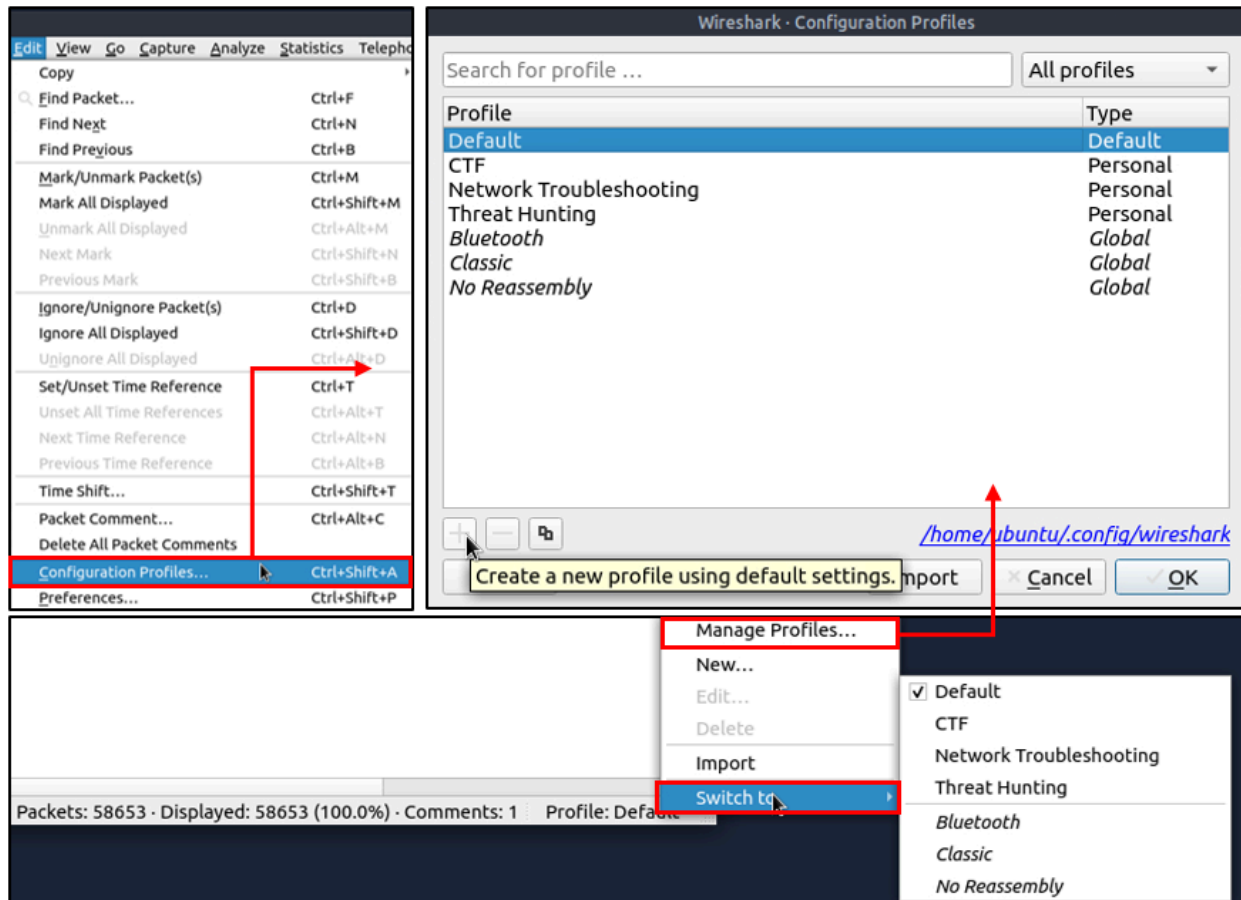
Bookmarks and Filtering Buttons: Allow analysts to save frequently used or complex Wireshark filters as bookmarks or buttons, enabling quick reuse and improving investigation efficiency.



Creating and using display filter buttons:



Profiles: Allow analysts to create and switch between different Wireshark configurations, including filters, coloring rules, and preferences, making it easier to adapt the tool for different investigation scenarios. "Edit → Configuration Profiles" menu or the "lower right bottom of the status bar → Profile"



Answer the questions below

Find all Microsoft IIS servers. What is the number of packets that did not originate from "port 80"?

21 (run the query: http.server contains "IIS" && !tcp.srcport == 80)

Find all Microsoft IIS servers. What is the number of packets that have "version 7.5"?

71 (run the query: http.server contains "IIS" and http.server matches "7.5")

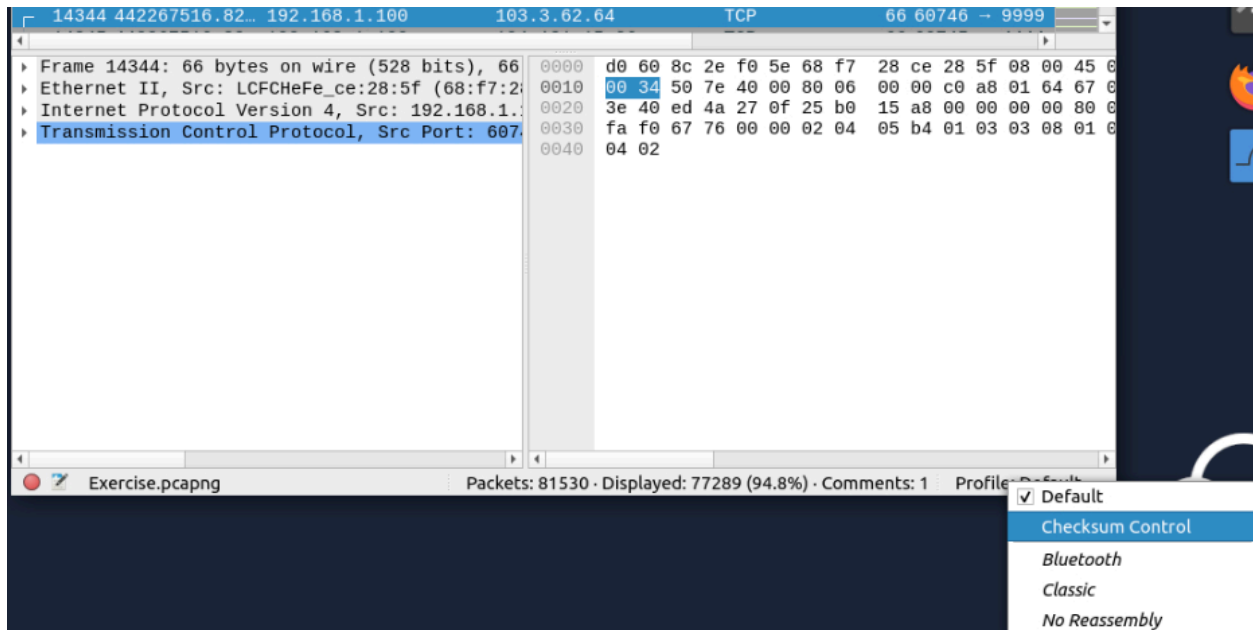
What is the total number of packets that use ports 3333, 4444 or 9999?

2235 (run the query: tcp.port in {3333 4444 9999})

What is the number of packets with "even TTL numbers"?

77289 (run the query: string(ip.ttl) matches "[24680]\$")

Change the profile to "Checksum Control". What is the number of "Bad TCP Checksum" packets? 34185



(then go analyse > display filter expression)

Wireshark · Display Filter Expression

Field Name	Relation
- IB · InfiniBand - TCP · Transmission Control Protocol - tcp.checksum · Checksum - tcp.checksum.ffff · TCP Checksum 0xffff inste... - tcp.checksum.status · Checksum Status - tcp.checksum_bad.expert · Bad checksum - tcp.checksum_calculated · Calculated Checks... - tcp.options.mptcp.checksum · Checksum - tcp.options.mptcp.checksumreq.flags · Check...	- is present == != > < >= <= - in

Value (Unsigned integer, 1 byte)

0

Predefined Values

- Bad**
- Good
- Unverified
- Not present

Range (offset:length)

Search: tcp.check

tcp.checksum.status == 0

Click OK to insert this filter

? Help

× Cancel

✓ OK

tcp.checksum.status == 0

No.	Time	Source	Destination	Protocol	Length	Info
14341	442267516.82...	192.168.1.100	101.201.172.235	TCP	66	60740 → 8888 [SYN]
14342	442267516.82...	192.168.1.100	103.3.62.64	TCP	66	60742 → 16666 [SYN]
14343	442267516.82...	192.168.1.100	103.3.62.64	TCP	66	60743 → 19999 [SYN]
14344	442267516.82...	192.168.1.100	103.3.62.64	TCP	66	60746 → 9999 [SYN]
14345	442267516.82...	192.168.1.100	104.131.15.86	TCP	66	60745 → 4444 [SYN]
14346	442267516.82...	192.168.1.100	103.3.62.64	TCP	66	60744 → 6666 [SYN]
14347	442267516.82...	192.168.1.100	101.201.172.235	TCP	66	60741 → 8889 [SYN]
14348	442267516.82...	192.168.1.100	104.140.244.186	TCP	66	60747 → 3333 [SYN]
14349	442267516.82...	192.168.1.100	104.140.244.186	TCP	66	60748 → 5555 [SYN]
14350	442267516.82...	192.168.1.100	104.140.244.186	TCP	66	60749 → 7777 [SYN]
14351	442267516.82...	192.168.1.100	104.140.244.186	TCP	66	60750 → 80 [SYN]
14352	442267516.82...	192.168.1.100	118.178.149.200	TCP	66	60751 → 3357 [SYN]
14353	442267516.82...	192.168.1.100	104.236.136.96	TCP	66	60752 → 3333 [SYN]
14354	442267516.82...	192.168.1.100	120.77.125.70	TCP	66	60753 → 3333 [SYN]
14355	442267516.82...	192.168.1.100	106.14.95.39	TCP	66	60754 → 8008 [SYN]
14356	442267516.82...	192.168.1.100	107.191.60.255	TCP	66	60755 → 3333 [SYN]

Frame 14344: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface
 Ethernet II, Src: LCfCHeFe_ce:28:5f (68:f7:28:ce:28:5f), Dst: zte_2e:f0:5e (d8:01:2e:f0:5e)
 Internet Protocol Version 4, Src: 192.168.1.100, Dst: 103.3.62.64
 Transmission Control Protocol, Src Port: 60746, Dst Port: 9999, Seq: 0, Len: 0

Exercise.pcapng Packets: 81530 · Displayed: 34185 (41.9%) · Comments: 1 Profile: Checksum Control

Use the existing filtering button to filter the traffic. What is the number of displayed packets? 261

(http.response.code == 200) && (http.content_type matches "image/gif|jpeg") gif/jpeg with http-200

No.	Time	Source	Destination	Protocol	Length	Info
10222	211538308.50...	65.54.95.140	192.168.3.131	HTTP	538	HTTP/1.1 200 OK
10238	211538308.51...	76.74.140.168	192.168.3.131	HTTP	585	HTTP/1.1 200 OK
10240	211538308.55...	65.54.95.140	192.168.3.131	HTTP	691	HTTP/1.1 200 OK
10245	211538308.57...	65.54.95.140	192.168.3.131	HTTP	276	HTTP/1.1 200 OK
10298	211538308.60...	65.54.95.140	192.168.3.131	HTTP	799	HTTP/1.1 200 OK
10303	211538308.60...	65.54.95.140	192.168.3.131	HTTP	890	HTTP/1.1 200 OK
10306	211538308.60...	65.54.95.66	192.168.3.131	HTTP	148	HTTP/1.1 200 OK
11559	211538348.97...	91.103.140.2	10.0.2.15	HTTP	323	HTTP/1.1 200 OK
11580	211538349.19...	96.17.8.49	10.0.2.15	HTTP	382	HTTP/1.1 200 OK
11852	211538355.33...	208.82.236.130	192.168.3.131	HTTP	480	HTTP/1.1 200 OK
12195	211538355.85...	209.17.73.30	192.168.3.131	HTTP	526	HTTP/1.1 200 OK
12259	211538355.90...	209.17.73.30	192.168.3.131	HTTP	1395	HTTP/1.1 200 OK
12309	211538356.68...	208.82.236.130	192.168.3.131	HTTP	246	HTTP/1.1 200 OK
12323	211538356.74...	208.82.236.130	192.168.3.131	HTTP	1437	HTTP/1.1 200 OK
13253	211538379.92...	70.37.129.32	10.0.2.15	HTTP	525	HTTP/1.1 200 OK
13555	211538394.91...	70.37.129.34	10.0.2.15	HTTP	261	HTTP/1.1 200 OK

▶ Frame 13555: 261 bytes on wire (2088 bits), 261 bytes captured (2088 bits) on
 ▶ Ethernet II, Src: RealtekU_12:35:02 (52:54:00:12:35:02), Dst: PcsCompu_cc:3f:3
 ▶ Internet Protocol Version 4, Src: 70.37.129.34, Dst: 10.0.2.15
 ▶ Transmission Control Protocol, Src Port: 5480, Dst Port: 2554, Seq: 6177, Ack: 6177
 ▶ [7 Reassembled TCP Segments (6383 bytes): #13546(1048), #13547(1460), #13549(6383)
 ▶ **Hypertext Transfer Protocol**
 ▶ JPEG File Interchange Format

0000 08 00 27 cc
 0010 00 f7 00 6d
 0020 02 0f 15 68
 0030 22 38 8c 34
 0040 24 61 79 08
 0050 91 c7 0a 47
 0060 bc c4 d7 11
 0070 9d f1 f9 57
 0080 0d 2c 87 7b
 0090 bb 73 c7 53
 00a0 02 df 5e 6a
 00b0 96 95 63 65
 00c0 cc b3 6d 6c
 00d0 cc b2 ac 91
 00e0 bc bc 6b 70
 00f0 05 2d 88 6d
 0100 46 57 3f ff

Frame (261 bytes)

Exercise.pcapng Packets: 81530 · Displayed: 261 (0.3%) · Comments: 1 Profile: Checksum Control